

**The Social Foundation of Trade Unions:
Social Capital, Union Participation and Wage Effect in the
United States**

Journal:	<i>Socio-Economic Review</i>
Manuscript ID	SER-2025-0593.R1
Manuscript Type:	Article
SER Keywords:	trade unions, social capital, collective bargaining, labor markets
JEL Classification :	J51- Trade Unions: Objectives, Structure, and Effects, Z13- Economic Sociology; Economic Anthropology; Social and Economic Stratification



1 Introduction

Trade unions have long been recognised as central institutions shaping wage inequality and labour market outcomes. Despite their recent decline across many advanced economies, particularly in the United States, a large body of research has documented their role in raising wages, compressing wage distributions, and protecting workers' welfare (Wallerstein, 1999; Wallerstein and Western, 2000; Card, Lemieux and Riddell, 2003, 2020; Western and Rosenfeld, 2011; Fortin, Lemieux and Lloyd, 2021; Firpo, Fortin and Lemieux, 2009; Farber et al., 2021; Reynolds and Brady, 2012; Rhodes, 2026; VanHeuvelen, Han and VanHeuvelen, 2025; VanHeuvelen and Brady, 2022; Brady, Baker and Finnigan, 2013; Jäger, Naidu and Schoefer, 2024).

An interdisciplinary literature has sought to explain trade unions' effects on labour market outcomes and the changes in unionisation across countries and over time. On the one hand, economic models have explained trade unions' effects on labour market outcomes through market mechanisms, such as the monopoly of labour supply and spillover effects (Freeman and Medoff, 1992; Fortin, Lemieux and Lloyd, 2021; Jäger, Naidu and Schoefer, 2024). On the other hand, the comparative and institutional literature has examined how institutional arrangements and political contexts affect unionisation and its effects across countries and over time by shaping the institutional and power constraints for labour organisation (Ebbinghaus and Visser, 1999; Western, 1993, 1994, 1995; Wallerstein and Western, 2000; Brady and Leicht, 2008; Jacobs and Myers, 2014; VanHeuvelen, 2020, 2023). Finally, sociological research has highlighted the embeddedness of labour market relations in broader social and cultural structures, pointing to extra-market mechanisms such as moral economy and collective activism, through which unions could shape labour market outcomes (Western and Rosenfeld, 2011; VanHeuvelen, 2018; Wilmers, 2017). Together, these perspectives have generated rich insights into how unions operate and how their prevalence and effects vary across contexts.

What these approaches leave less well specified, however, is the micro-social founda-

tion of trade unions as collective action. Economic and institutional accounts explain the mechanisms and constraints of union collective action, but they say less about how workers actually overcome the coordination and free-rider problems inherent to all collective action, especially given market and institutional barriers. While sociological accounts have emphasised the embeddedness of trade unions in socio-cultural structures, how such embeddedness enables union organisation and extra-market mechanisms remains under-theorised. Furthermore, sociological claims about the role of extra-market factors and mechanisms in shaping labour market relationships have been empirically underpowered by research designs that use indicators with ambiguous interpretations that are also compatible with market-based explanations.

This study complements the existing literature by examining civil society as the infrastructural condition that embeds and enables union collective action. Drawing on theories of collective action, social networks and social capital, I conceptualise unionisation and bargaining as collective action processes embedded in local social structure and use the concept of social capital to describe three distinct features of civil society that may enable union collective action. The first is economic connectedness, which captures individuals' access to external resources and cross-cutting ties in the community. The second is network clustering, which captures social closures that facilitate cooperation and diffusion. The third is civic engagement, which captures the overall preferences and norms for participation and the prevalence of civic engagement in a community.

Empirically, rather than relying on conventional survey proxies or broad aggregate indicators of civic activity, the study uses a novel dataset of multiple county-level measures of distinct forms of social capital derived from the Facebook friendship networks of millions of users and digital traces of civic organisations in the United States (Chetty et al., 2022). This dataset allows a more direct and systematic empirical test of distinct network-based and cultural mechanisms that was previously infeasible. Linking these measures to individual-level data from the Current Population Survey between 2013 and 2019, estimates from multi-

level models show that higher levels of network clustering are consistently associated with a higher probability of union participation and a stronger union wage effect. On average, a 1 standard deviation increase in the county's level of network clustering is associated with a 2.3 percentage point increase in workers' probability of participation in trade unions and a 2% increase in the union wage premium. The association with union participation is also stronger in counties with higher levels of civic engagement. In contrast, evidence for economic connectedness and civic engagement is weaker and more inconsistent.

The study contributes to the interdisciplinary literature on trade unions and labour market relationships in three ways. First, theoretically, it provides an account for the micro-social foundation of trade union collective action in civil society, bridging literature in labour market relationships, civil society, social capital and collective action. Second, empirically, it offers novel evidence on the role of extra-market factors in shaping labour market relationships and outcomes. Finally, using multidimensional measures of social capital, the study systematically tests the effects of different types of social features on union collective action and identifies network clustering as the most important positive factor. It therefore sheds light on the specific social conditions that enable union collective action.

Taken together, the study suggests that understanding the decline and persistence of unions, and their consequences for labour market outcomes, requires attention not only to institutional and market conditions but also to the local social structure that makes labour organisation and activism feasible. In turn, it implies that labour democracy could, to some extent, be rebuilt from the bottom up, through strengthening social ties and community-based organisation, rather than relying solely on institutional changes.

The rest of the article is arranged as follows: Section 2 reviews the interdisciplinary literature on trade unions and labour market relationships and introduces civil society as a complementary perspective. Section 3 introduces the three types of social capital; outlines the mechanisms through which they affect union collective action, and derives the empirical hypotheses. Section 4 describes the empirical research design and defines the theoretical

estimands. Sections 5 and 6 describe the data, measurements and empirical methods. Section 7 presents empirical results, and Section 8 discusses the study's limitations and concludes with implications for the literature and future research.

2 Theoretical Motivations

2.1 Explanations of Unionisation and Collective Bargaining Power

Trade unions have been widely recognised as a key labour market institution in developed economies. Despite their general decline across countries in recent decades, an interdisciplinary body of literature has extensively documented the positive effects of trade union membership and unionisation levels on workers' outcomes, such as income, health, and wealth, and the mitigating effects on income inequality and working poverty (Wallerstein, 1999; Wallerstein and Western, 2000; Card, Lemieux and Riddell, 2003, 2020; Western and Rosenfeld, 2011; Fortin, Lemieux and Lloyd, 2021; Firpo, Fortin and Lemieux, 2009; Farber et al., 2021; Reynolds and Brady, 2012; Rhodes, 2026; VanHeuvelen, Han and VanHeuvelen, 2025; VanHeuvelen and Brady, 2022; Brady, Baker and Finnigan, 2013; Jäger, Naidu and Schoefer, 2024). How do unions acquire and sustain such power, and why has union strength declined in many contexts? Scholars in economics, comparative political economy, and sociology have approached these questions from different analytical perspectives. The following section reviews these approaches, their contributions and limitations.

2.1.1 Economic Models

In the standard microeconomic labour market model, trade unions are conceptualised as utility-maximising producers of labour supply with monopolistic power from collective bargaining (Jäger, Naidu and Schoefer, 2024). With this monopolistic power, unions could set the cost of labour (i.e., wage and non-wage benefits) above the market-clearing level (Freeman and Medoff, 1992). In a perfectly competitive market, this would lead to inefficiencies

and job losses. However, in the presence of monopsony - where firms can set wages below the competitive level - union monopoly may restore wages toward equilibrium and redistribute rents from firms to workers, potentially improving both efficiency and worker welfare.¹ More recent research has also modelled and shown that such wage-setting power could have positive spillover effects on non-unionised workers' wages, and thus market-wide wage levels through threats of potential unionisation (Fortin, Lemieux and Lloyd, 2021).

Given unions' positive effects on workers' market position and bargaining power, it is predicted that utility-maximising workers would choose to unionise if the relative benefits outweigh the costs of organisation and potential efficiency loss. The decline in unionisation has therefore been attributed to changes in the relative costs and benefits of organising — for example, shifts in macroeconomic conditions, firm productivity and profitability, and workforce composition across industries and demographic groups (Western, 1994; Dickens and Leonard, 1985; Bryson and Blanchflower, 2009; Schaller, 2023).

2.1.2 Comparative Political Economy and Institutional Analysis

Research on comparative political economy presents a more nuanced picture by highlighting the role of institutional contexts in shaping unions' presence and bargaining power through the distribution of power resources among different actors in labour relationships (Korpi, 1985; Brady and Sosnaud, 2010). Comparative studies have documented substantial variation in legislation and the contents and modes of union organisation and collective bargaining across developed countries (see e.g., Bhuller et al., 2022; Jäger, Naidu and Schoefer, 2024, for reviews). For example, in the so-called “Wagner Act” system in the United States and Canada, union representations are established and limited at the workplace level; whereas in European countries like France or Nordic countries with the “Ghent system”, industry and

¹The presence of labour market monopsony and the effects of unions on employment and productivity are subjects of continuing debates in labour economics. See, for example, Barth, Bryson and Dale-Olsen (2020) and Alder, Lagakos and Ohanian (2023) for contrasting evidence on the effect of unions on economic performance. See also Card (2022), Manning (2021), Yeh, Macaluso and Hershbein (2022) for discussions on monopsony.

national level union representations are more prevalent, and unions are tied to the national welfare system. Empirical research has shown that these institutional factors could explain the cross-national and temporal variations in the levels and rates of change in unionisation, and also moderate the effect of structural factors (Ebbinghaus and Visser, 1999; Western, 1993, 1995, 1994; Wallerstein and Western, 2000).

Along a similar line, research has also found that within countries, unionisation and its effects on workers' welfare are contingent on the political context. For example, research has found that the rise of right-wing politics in the United States and other Western economies since the 1980s has significantly undermined unions' presence and effects on the income distribution (Western, 1995; Brady and Leicht, 2008; Jacobs and Myers, 2014). Relatedly, Massenkoff and Wilmers (2024) find that the positive wage effect of strikes in the United States disappeared after 1981. In two recent studies, VanHeuvelen (2020, 2023) demonstrates that in the United States, state-level adoption of the Right-to-Work (RTW) law removed the negative associations between union density and income inequality, particularly in states with a higher presence of unions; and demobilised unions' popularity and bargaining power in both the short and long runs. From this political economy perspective, unions and their power are shaped by institutional and political contexts rather than emerging solely from market forces.

2.1.3 Sociological Approach

Based on the power resource theory, while the institutional approach has focused on the macro-level context of power resource distribution, sociological research has highlighted the role of socio-cultural power resources and related extra-market mechanisms in shaping labour market outcomes.

For example, Western and Rosenfeld (2011) argue that trade unions could affect firms' wage-setting practices and reduce income inequality by institutionalising a "moral economy" of egalitarian wage norms (see also VanHeuvelen, 2018). Following this perspective, Wilmers

(2017) develops a more general framework that conceptualises unions as activist groups. He argues that through collective activism and campaigns, trade unions could achieve collective bargaining outcomes and improve workers' welfare without directly changing their market positions, through, for example, strikes. In this way, the sociological approach provides a different explanatory perspective to the market-centred economic models.

2.2 Limitations of Existing Explanations

In sum, existing research in economics, comparative political economy and sociology has painted a rich picture of how trade unions organise and affect labour market outcomes through market and extra-market mechanisms, and how such processes are contingent on the institutional and political contexts. However, despite these contributions, two issues remain underdeveloped.

2.2.1 Where Do Unions Come From?

The first ambiguity concerns the question: where do unions come from? While treating trade unions as analytical subjects, much of the literature remains less explicit about how unionisation and its effects emerge from workers' collective action.

For example, the economic model predicts that utility-maximising workers would naturally unionise if the returns are positive. However, unionisation, like all collective behaviour, requires coordinated actions in the face of the free-rider problem (Olson, 2009). While some research has suggested economic and sociological mechanisms through which unions overcome the problem, such arguments have not been systematically incorporated in the literature nor widely tested (Booth, 1985; Murphy, 2020).

Similarly, while arguing for the effects of trade unions as moral economy institutions (Western and Rosenfeld, 2011) or activist groups (Wilmers, 2017), sociological research has not explained why such societal collective effects are possible. For instance, it is not self-evident why trade unions could effectively exert normative pressure and why employers

would comply with such pressure. Finally, although the institutional literature explains macro-structural and institutional conditions for union organisation, strength and decline, it does not specify the micro-level social conditions that enable union mobilisation.

In sum, the micro-foundations of union organisation and action remain largely under-specified in the existing literature, leaving the theoretical account of how unionisation and its labour market effects operate incomplete.

2.2.2 Market versus Extra-market Mechanisms

The second ambiguity concerns more specifically the sociological arguments and evidence for the extra-market mechanisms of trade unions. Sociological research has suggested non-market mechanisms as complementary - if not alternative - explanations to the economic market model. For example, Wilmers (2017) describes collective activism and non-market power as alternative explanations to the unions' wage-setting power in the context of weak union market power. Empirically, however, these mechanisms are often operationalised using indicators that remain equally compatible with market-based explanations, thus making it difficult to interpret the results as evidence for the role of extra-market mechanisms.

For example, to test unions' norm-enforcing effect on wage setting, Western and Rosenfeld (2011) and VanHeuvelen (2018) use region-industry-specific union density as the proxy for unions' power and interpret the negative relationship between union density and overall wage inequality as evidence supporting the norm enforcement explanation. However, the empirical relationship could be explained by the conventional economic wage distribution and union spillover model (Jäger, Naidu and Schoefer, 2024; Fortin, Lemieux and Lloyd, 2021). Similarly, Wilmers (2017) uses union spending as the measure of unions' investment in activism. Despite careful efforts to address endogeneity, the positive association between union spending and the union wage effect can still be interpreted as reflecting market mechanisms, since investments in activism and campaigns could be seen as investments in and manifestations of unions' monopolistic market power.

In these cases, the empirical indicators and relationships fail to clearly distinguish between sociological and economic mechanisms to support the proposed importance of extra-market social factors. This leads to what Abbott (1997) describes as "narrative ambiguity", where the empirical evidence can be plausibly interpreted through multiple theoretical explanations, and thus cannot effectively support the proposed explanation.

2.3 Civil Society as the Missing Link

In this study, I argue that civil society - as the infrastructural social conditions in which union organisation and actions are embedded - provides a critical link that complements existing research.

First, civil society factors help explain the emergence of trade unions as collective organisations and labour market institutions. At the micro-behavioural level, features such as social connections, norms and trust reduce the cost of collective action by easing coordination and mitigating the free-rider problem (Dickens and Leonard, 1985). Indeed, case studies on union organisation and renewal have documented the importance of existing communities and relationships as the basis of effective unionisation (Fine, 2005; Nissen and Jarley, 2005; Hyman, 2007; Saundry, Stuart and Antcliff, 2012; Tapia, Ibsen and Kochan, 2015; Murray, 2017). At the historical-institutional level, voluntary associations and solidarity have been important in shaping the development of labour market and state institutions. For example, in his classic account of the development of the Swedish union movement and welfare state, Korpi (1978) highlights the roles of workplace and community relationships as the basis of worker organisation and struggle. Quantitative cross-sectional analysis by Ebbinghaus, Göbel and Koos (2011) also shows that the density of civic organisation, in addition to economic and institutional factors, can explain variations in union density among European countries. Overall, these studies point to the important role that civil society plays in union organisation, yet it has been much less systematically incorporated as a central explanatory dimension alongside institutional and market factors.

Second, civil-society factors could serve as enabling factors to extra-market mechanisms of union organisation and collective bargaining. For example, unions need a strong civil society to effectively exert normative pressure (Western and Rosenfeld, 2011) or mobilise collective activism (Wilmers, 2017). Through such mechanisms, civil society could shape unions' bargaining power and thus workers' outcomes, independently of market mechanisms.

In these ways, civil society helps address the two limitations described above. First, it provides micro-foundations for the emergence of trade union collective action through mechanisms such as social diffusion, trust and control. Second, by enabling collective action and bargaining, civil society factors could shape labour market outcomes *prior to* market mechanisms. This creates leverage to directly test the effect of extra-market factors on labour market relationships and outcomes, without relying on market indicators that risk conflation with economic explanations.

In sum, civil society presents an analytical perspective for understanding how union collective action is embedded in and enabled by social structural factors. The next section specifies the mechanisms through which civil society may shape union organisation and bargaining power.

3 Mechanisms and Hypotheses

I use the concept of social capital to summarise the various civil society factors. Broadly defined, it refers to features of the social structure that are productive to collective action, derived from individuals' social relationships and participation (Portes, 1998). Here, I use the concept to capture social conditions — such as resource availability, network structures, and participatory norms — that enable cooperation and, in turn, facilitate union organisation and collective action.

One major limitation of the social capital concept, however, is that it has multiple definitions and implied mechanisms, which create inconsistencies in the literature (see e.g., Portes,

1998; Jackman and Miller, 1998; Lin, 2001). The definitions vary even in the relatively small literature that uses the concept to explain union organisation. For example, Hyman (2007)'s definition of social capital focuses on individual workers' union-based connections; Saundry, Stuart and Antcliff (2012)'s study focuses on how existing social connections among workers provide grounds for union activism; and Ebbinghaus, Göbel and Koos (2011) treat social capital as a country/society's macro-level cultural property, measured by the average density of civil participation. Such inconsistency makes it difficult to systematically evaluate the role of civil society and social capital in shaping union organisation.

Underlying these inconsistent definitions are different social features and mechanisms that enable collective action. In contrast to existing studies that treat social capital as a single undifferentiated concept, in this study, I examine three types of social capital defined at different levels, and the mechanisms through which they affect union collective action: namely, micro-level individual connections, meso-level interpersonal network structure, and macro-level culture of civic participation.

3.1 The Micro Level: Resource and Solidarity through Social Connections

At the micro-level, social capital refers to individuals' social ties that can bring productive resources and information. Extensive research in sociology has documented its positive effect on individuals' educational and labour market outcomes (Bourdieu, 1986; Coleman, 1988; Granovetter, 2018; Lin, Ensel and Vaughn, 1981).

In the context of union collective action, workers with more extensive, resourceful social ties are better positioned to contribute to the collective action, either directly or by attracting external support. Aggregated across individuals, higher levels of such capital within a community can expand unions' resource bases, enabling broader mobilisation and more effective campaigns and bargaining. This logic is consistent with the resource mobilisation theory in social movements, which emphasises the critical role of financial, organisational,

and informational support from networked actors (Edwards and McCarthy, 2007).

Besides direct resource channelling, social ties, especially those cross-cutting different social groups, could also enable individual and collective action by creating wider solidarity and support. For example, Gould (1991, 1993*b*) found that neighbourhood-based networks enabled mobilisation across class and trades during the Paris Commune. A similar mechanism could also apply to union mobilisation as unions draw support through cross-cutting social ties in the community.

Therefore, union collective action should be more effective in communities where members have more resourceful and cross-cutting social ties. Thus, I derive Hypothesis 1:

H1 : The effectiveness of union collective action is positively associated with the level of economic connectedness in the community.

3.2 The Meso Level: Intragroup Controls and Diffusion in Transitive Networks

At the meso interpersonal level, social capital refers to the structural features of interpersonal networks that shape group dynamics. Extensive research in social movement studies has examined the role of social networks in collective action through mechanisms such as information diffusion and action coordination (Gould, 1993*a*; Diani and McAdam, 2003; Diani and Mische, 2015). In this study, I focus on one particular structural feature: transitivity, defined as the prevalence of closed triads in a network. I argue that transitive networks strengthen union workers' collective action through two mechanisms: intragroup control and social influence.

The control mechanism — highlighted in Coleman (1988)'s analysis of the Jewish diamond trade (see also Richman, 2002, 2006) — suggests that dense, closed networks reinforce social norms and cooperation through reputation, mutual monitoring and sanctioning for defections. Therefore, transitive networks can enable union collective action by increasing

the cost of free-riding, thus mitigating the free-rider problem (Olson, 2009; Hardin, 1971; Heckathorn, 1988).

Transitive network structures can also strengthen collective action through the diffusion mechanism. It is well established that people are more likely to act when those around them do so, due to sub-mechanisms such as information transfer, updates on perceived cost and benefits, and direct influences (e.g. McClurg, 2003; Granovetter, 1978), leading to the diffusion of ideas and actions. Transitive networks could make the diffusion process more efficient as individuals receive and exert stronger and more frequent influences through dense ties. Using network simulations, Siegel (2009) finds that the density of transitive triads in networks, measured by the global clustering coefficient, is important in explaining variations in participation rates in different network structures. He describes the densely connected clusters as “enclaves of participation” in which the participatory behaviour of one individual can quickly affect others through dense ties, thereby initiating the diffusion of participation. This facilitating effect of closed networks is particularly important for what Centola and Macy (2007) describe as “complex contagion”, which involves the diffusion of complex information and costly actions that require repeated interactions. Trade union collective action, which often requires persuading workers to participate in union elections, strikes or campaigns, is a case of complex contagion that benefits from dense transitive networks for effective diffusion.

Therefore, taken together, through intra-group control and influence mechanisms, I derive Hypothesis 2:

H2 : The effectiveness of union collective action is positively associated with the level of network clustering in the community.

3.3 The Macro Level: Civic Engagement and Participatory Preferences

While micro- and meso-level social capital concern personal and interpersonal network connections, macro-level social capital is conceptualised as the overall characteristic of a community, capturing the prevailing preferences and norms on participation and the prevalence of civic engagement and cooperation. Popularised by the works of Putnam (1994, 2000) in political science, this type of social capital has been viewed and examined as the socio-cultural basis for political cooperation, democracy and economic developments (e.g. Fukuyama, 1995, 2001; Inglehart, 1997). In sociology, Sampson (2013) proposes the similar concept of “collective efficacy” as an explanatory factor for neighbourhood differences in the level of social control and crime.

From a collective action perspective, the level of civic engagement reflects individuals’ internal propensity to participate. Individuals in communities with higher levels of civic engagement, on average, are expected to be more willing to cooperate and participate in collective action. In Granovetter (1978)’s threshold model, this implies lower and less dispersed participation thresholds. The inherent cost of collective action is therefore lower in communities with higher levels of civic engagement.

Civic engagement can also strengthen union collective action by providing the cultural and organisational basis. For example, an active civil society may function as a source of normative power that provides legitimacy for unions’ normative pressure and activism (Reuter, Wijkström and Meyer, 2014). Furthermore, a more active civil society could empower unions by providing the organisational basis for mobilisation, coordination and alliance building (Dixon and Martin, 2012).

Through these mechanisms, civic engagement could directly enable union collective action. Therefore, I derive Hypothesis 3:

H3: The effectiveness of union collective action is positively associated with the level of

civic engagement in the community.

Civic engagement can also affect collective action indirectly by moderating the effect of network clustering. First, when participation thresholds are lower, the network diffusion process becomes more efficient because individuals are more likely to initiate actions, either autonomously or due to social influences. Studies on networks and collective action have shown that the network diffusion of participation is contingent on individuals' preferences (Gould, 1993a; Siegel, 2009). If people are reluctant to make initial contributions, as in what Siegel (2009) described as the "weak motivation class", then the rate and the final level of the spread of participation will be limited even with favourable network structures. Conversely, the effect of the initial contribution of high-motivation individuals on the population participation rate could be amplified by network structures, particularly when the initial preferences of others are also high.

Second, civic engagement may strengthen the control mechanism by mitigating what Heckathorn (1989) described as the "second-order free-rider problem", in which case the control mechanism faces the same problem of collective action as people may not contribute to the enforcement of norms. As in the first-order free-rider problem, a higher level of civic engagement can mitigate the problem as people have a greater tendency to enforce social norms, thereby strengthening the effect of network clustering on norm enforcement and cooperation.

These mechanisms suggest positive interactions between network structures and individuals' preferences for participation in facilitating collective action. Thus, I derive Hypothesis 4:

H4 : The effect of network clustering on union collective action is positively associated with the level of civic engagement in the community.

4 Research Design

This study examines the effect of three types of social capital on union collective action. Empirically, the study focuses on the United States labour market between 2013 and 2019 and two outcomes linked to union collective action: union participation and the union wage effect.

4.1 The United States Context

The United States presents an informative case for examining the relationship between civil society and trade unions, given its decentralised labour market institutions and long-standing emphasis on civic association.

Institutionally, the United States labour market is characterised by a limited welfare state and a decentralised, workplace-based system of collective bargaining (Bhuller et al., 2022; Jäger, Naidu and Schoefer, 2024). Moreover, Right-to-Work laws in many states allow workers in union-represented workplaces to remain non-members while enjoying the benefits of representation, effectively institutionalising free-riding and increasing the costs of unionisation and collective action (VanHeuvelen, 2020, 2023). In contrast to coordinated market economies with sectoral bargaining and stronger state guarantees, the United States system therefore places greater organisational burdens on workers themselves, thereby making the social preconditions of mobilisation more visible.

At the same time, the United States has historically been characterised by a dense associational life and a strong normative emphasis on civil society. This strong civil society has often been viewed as a distinctive strength of American democracy and, at times, even as a functional substitute for more expansive state institutions (e.g., Putnam, 2000, see also Skocpol (1997) for a more critical view).

Taken together, this combination of decentralised labour institutions, heightened collective action constraints, and the historical emphasis on civil society makes the United States

a theoretically useful setting for examining the relationship between social capital and union organisation and bargaining power.

The time frame for the study is chosen for substantive and practical reasons. Substantively, the period covers two administrations - the second Obama administration and the first Trump administration - with contrasting economic policy agendas. Previous studies have shown that unions' bargaining power was affected by the rise of right-wing politics (Brady and Leicht, 2008; Jacobs and Myers, 2014). The political change in this period, therefore, provides an informative case study for examining the robustness and heterogeneity of the effects in different socio-political contexts. The study, furthermore, focuses on the pre-COVID-19 era to avoid the period effects of economic shocks, and thus does not include the year 2020. Methodologically, the time frame is short and close to 2020, so that the social capital measures - as explained later - could be plausibly assumed as constant. The following sections describe the empirical approach in more detail.

4.2 Outcome Measures and the Theoretical Estimands

To examine the effect of social capital on union collective action, I focus on two outcomes: workers' union participation and the union wage effect. Worker participation is a prerequisite for effective union organisation and collective bargaining. It is widely used as an indicator of organised labour power in within-country and comparative studies (e.g. Western, 1995; Western and Rosenfeld, 2011; VanHeuvelen, 2018; Ebbinghaus, Göbel and Koos, 2011). In the United States context, where workplace-level union representation requires support from the majority of workers, worker participation is fundamentally linked to the success and effectiveness of union collective action.

While union participation captures unions' organisational success and potential bargaining power, the union wage effect captures unions' realised bargaining effect on one of the most important labour market outcomes. Although research has highlighted unions' effect on non-monetary outcomes such as working conditions and security, wages remain the most

central and widely studied outcome in economics and sociology (e.g. VanHeuvelen, 2018, 2020, 2023; Wilmers, 2017; Farber et al., 2021).

Although many existing studies have examined union collective action using aggregate-level measures such as country, region or industry-level union density (e.g., Western, 1995, 1993; Western and Rosenfeld, 2011; Ebbinghaus, Göbel and Koos, 2011), such analysis may be subject to ecological fallacy and unobserved worker-level selection bias and heterogeneity (VanHeuvelen, 2018). Therefore, I examine the two outcomes at the individual level. The theoretical counterfactual estimands are (Lundberg, Johnson and Stewart, 2021):

(1) The average treatment effects (ATEs) of community j 's social capitals on worker i 's propensity of union participation:

$$\frac{1}{N} \sum_t \sum_j \sum_i [P(Union_{ijt} = 1 \mid \mathbf{SC}_{jt} = \mathbf{sc}) - P(Union_{ijt} = 1 \mid \mathbf{SC}_{jt} = \mathbf{sc}')] \quad (1)$$

where $\mathbf{SC}_{jt}$ is a vector of measures of county j 's micro-, meso- and macro-level social capital in year t ,

and,

(2) The conditional average treatment effect (CATE) of union membership on wage, conditional on the level of social capital:

$$\frac{1}{N_{sc}} \sum_t \sum_{j: \mathbf{SC}_j = \mathbf{sc}} \sum_i [Wage_{ijt} (Union = 1) - Wage_{ijt} (Union = 0)] \quad (2)$$

The following sections describe the empirical approach in more detail.

5 Data and Measurements

5.1 Social Capital Measurements

One of the main challenges in the quantitative analysis of social capital is measurement. Existing studies following the macro-level social capital approach have constructed granular county-level measures of civic engagement based on indicators such as the number of public facilities, civic organisations, and voting turnout (Rupasingha, Goetz and Freshwater, 2006; Vâlsan, Goschin and Druică, 2023; Peluso et al., 2024). However, some have questioned the conceptual and methodological validity of such measures, as they do not directly capture social networks or preferences and are subject to endogeneity problems (Jackman and Miller, 1998; Durlauf, 2002; Lin, 2001). Direct measures of social connections and network structures require mapping the entire social network structure and are extremely difficult to obtain for large geographic units from conventional data sources.

Recent advances in digital data offer the opportunity to examine social capital rooted in complete social networks. This study uses the cross-sectional county-level social capital measures developed by Chetty et al. (2022), based on the network data of 21 billion Facebook friendships among 72.2 million users aged 25–44 in 3090 counties in the United States in 2020. As part of Opportunity Insights, the dataset was initially collected to examine the causes of differences in neighbourhood effects on upward mobility in the US (Chetty et al., 2022). The dataset is publicly available at The Social Capital Atlas.

Based on county-level measures of the friendship network structure and other digital data, Chetty et al. (2022) construct multiple measures that capture different forms of social capital and examine their relationships with counties' effects on upward mobility. These measures are only weakly correlated with the existing measure by Rupasingha, Goetz and Freshwater (2006), suggesting that they capture distinct aspects of social capital.

Overall, derived from the friendship network structures, these measures allow more direct and systematic examinations of the relationship between social structural features and union

collective action outcomes. For this study, I use four county-level statistics: economic connectedness (EC), network clustering, volunteering rate, and density of civic organisations, to construct measures for micro-, meso-, and macro-level social capital. Below, I briefly explain the measures' methodology and links to the concepts.

5.1.1 Micro-level Social Capital: Economic Connectedness

As explained above, micro-level social capital captures the effect of individuals' social ties on collective action through direct resource channelling and building cross-cutting support. I use Chetty et al. (2022)'s economic connectedness (EC) to capture this form of social capital.

In Chetty et al. (2022), EC is first defined as an individual-level characteristic. A person's EC with individuals in a certain socioeconomic status (SES) quantile is defined as the proportion of friends belonging to that quantile. For example, in a two-quantile setting (i.e., below and above the median), the EC for a person in the lower SES-quantile with four friends, among whom one is in the upper quantile, is 0.25. Using two SES quantiles, Chetty et al. (2022) defines a county's EC as the average EC of people in the lower quantile.

Conceptually, EC captures individuals' ability, especially those in more disadvantaged positions, to access greater social resources through connections with advantaged individuals. It also captures the prevalence of cross-SES ties as the basis of cross-cutting solidarity and support. EC therefore provides an operational measure of micro-level social capital.

5.1.2 Meso-level Social Capital: Network Clustering

Meso-level social capital captures the prevalence of transitive network triads that enable the control and diffusion mechanisms. In network analysis, the presence of transitive triads can be measured by the local clustering coefficient, C_i . The local clustering coefficient, C_i , of a node i (i.e. individual) measures the proportion of closed triads in its ego-centred networks; that is, the proportion of a node's neighbour pairs that are themselves connected. (see Newman, 2018, Chapter 7.9.1). For example, for a person with three friends (A, B, C),

among whom two (A, B) are friends with each other, the person's local clustering coefficient is one-third.

For each county, Chetty et al. (2022) computes the network clustering level as the average local clustering coefficient across all individuals. I use this quantity as the measure of counties' network transitivity, as it provides a straightforward measure of the prevalence of transitive ties that facilitate control and diffusion mechanisms.²

5.1.3 Macro-level Social Capital: Civic Engagement Index

Macro-level social capital captures a community's overall level of civic engagement and participatory preferences. To measure counties' level of civic engagement, Chetty et al. (2022) use two statistics: the density of Facebook Pages of civic organisations and the share of people who are affiliated with at least one volunteering or activism Facebook group.

Civic organisations and volunteering rates have been widely used to construct an index of social capital to indicate the prevalence of civic participatory culture (e.g., Putnam, 2000; Rupasingha, Goetz and Freshwater, 2006). In this study, I follow this convention and construct an equal-weight index of civic engagement as the measure of civic engagement.

Such measures, however, have been criticised for risking tautological reasoning about the relationship between participatory culture and participation outcomes (e.g. Jackman and Miller, 1998; Portes, 1998). Further discussion of these conceptual issues is beyond the scope of this study, but the results should be interpreted with these caveats in mind.

5.1.4 Limitations

Before proceeding, some limitations of the data should be acknowledged and discussed. First, the measures are derived from a selected population of Facebook users aged 25-44, and their

²Another commonly used measure of network clustering is the global clustering coefficient C , calculated as the proportion of the total number of closed triads in a network to the number of all possible triads. This quantity is not computed or reported by Chetty et al. (2022). However, for the conceptualisation of meso-level social capital that focuses on interpersonal mechanisms and their effects on individual actions, the local clustering coefficient is arguably the more appropriate measure.

representativeness for the overall population and union workers may be limited. Nonetheless, Chetty et al. (2022) benchmark their data to the nationally representative survey data and find that the distributions of socio-demographic characteristics in the age range are overall consistent, thus validating the data's population representativeness. Furthermore, Blanchflower and Bryson (2022) estimate that the union density and propensity for union membership in the United States have inverse-U shape distributions by age that peak at 44-50. Therefore, while the age coverage of the Facebook data is limited, it includes at least half of the union member population, who may also be more responsive to mechanisms such as social influence.

Second, because the social capital measures are cross-sectional, they are assumed to be time-constant (i.e., $\mathbf{SC}_{jt} = \mathbf{SC}_j$) when applied to cross-year data. This assumption could be violated, introducing measurement errors. However, because counties' social network structures and civic culture are likely rooted in local social structures and cultures that change slowly, the errors are likely to be minor in the short run.

5.2 The Current Population Survey

The main analytical sample comes from the Merged Outgoing Rotation Group (MORG) data from the Current Population Survey (CPS) between 2013 and 2019. CPS uses a rotation group system in which each household is interviewed monthly in two four-month windows in a 12-month cycle. Weekly working hours and earnings are asked only in the final month of the two windows and the MORG dataset includes data collected from these final months of the year. Hourly earnings can be computed in the MORG data by dividing weekly earnings by working hours. The dataset also contains a range of respondents' demographic, socioeconomic, and employment information, including their union membership and coverage status.

In this study, I use MORG data pooled between 2013 and 2019. For data from each year, I include full-time employed non-agricultural workers aged 20-65 with valid union

membership information and county identifiers. I exclude samples with zero working hours and top- and bottom-code logged hourly earnings using the 99th and 1st percentiles. To avoid estimation bias in union wage premiums arising from data imputation for non-respondents in the CPS, I exclude samples with imputed earnings data (Hirsch and Schumacher, 2004). The final analytical data have a hierarchical structure where individual observations are nested in county-years. After excluding county-years with fewer than 5 observations, the final dataset consists of 196,728 individual observations nested in 2,068 county-years. Detailed distributions of observations by year and county are shown in Appendix A.

It should be noted that, because the primary purpose of CPS is to provide national and state-level estimates and to protect privacy, the availability of county identification is limited and county-level population inference is not available. Specifically, geographic identifications of small counties with populations less than 250,000 are synthesised or suppressed for privacy protection (U.S. Census Bureau, 2022). Therefore, including only observations with valid county identification creates a smaller selected sample of observations living in relatively populous areas. Descriptive statistics (see Table A1), however, show that the characteristics of the final selected samples are broadly similar to the original samples.

5.3 Additional Data

I include a range of additional county-level variables as covariates. For each year, I include county-level data on total population and the share of the white population, sourced from the United States Census estimates; GDP and the share of the government sector, sourced from the United States Bureau of Economic Analysis; unemployment rate, sourced from the Local Area Unemployment Statistics; the Republican's vote share in the last presidential election, sourced from the MIT Election Data+Science Lab. I also include the state-level Right-to-Work law status. Some county-years are missing in these datasets.

6 Methods

6.1 Assumptions and Models

To identify Estimands 1 and 2 from pooled cross-sectional observational data, the study relies on the unconfoundedness assumption, in addition to the positivity and consistency assumptions. While the positivity and consistency assumptions are likely to hold in this setting, the unconfoundedness assumption requires all confounders to be adjusted (Morgan and Winship, 2014).

At the individual level, I adjust for individuals' demographic characteristics using CPS-MORG data on respondents' sex, age, race and ethnicity. I also use CPS-MORG data on individuals' highest level of education and marital status. For their employment characteristics, I adjust for individuals' occupation, industry and sector (i.e., public versus private). For the county-year level, I adjust for demographic characteristics using the logged total population and the logged proportion of the white population. I adjust for macroeconomic conditions using the logged unemployment rate, logged GDP, and the logged GDP contribution by the government sector. For the political context, I adjust for the proportion of votes to the Republican party in the last general election and the RTW law status in the state. The assumed causal structure is discussed in more detail in Appendix B, and details of variable processing are described in Appendix C.

To estimate Estimand 1, I regress individuals' union membership status on the three social capitals, adjusting for covariates described above, in multilevel linear probability models with random intercepts:

$$P(Union_{ijt} = 1) = \beta_0 + u_{0jt} + \beta_1 \mathbf{SC}_j + \beta_2 \mathbf{X}_{ijt} + \beta_3 \mathbf{G}_{jt} + e_{ijt} \quad (1)$$

where, $\mathbf{SC}_j$ are the social capital variables and - optionally - their interactions; $\mathbf{X}_{ijt}$ are individual-level covariates; and $\mathbf{G}_{jt}$ are group-level covariates. u_{0jt} represents the normally distributed county-year-specific random intercepts, capturing county-year-specific levels of

union participation (Gelman and Hill, 2006). The parameter of interest is the vector of coefficients β_1 , which, under the identification assumptions, captures the effect of counties' social capital levels on individuals' propensity to join trade unions in Estimand 1.

To estimate Estimand 2, I regress individuals' logged hourly wages on their union membership status, interacted with the social capital variables, adjusting for the three social capitals and observed covariates, in multilevel linear models with random intercepts and slopes for union membership (Heisig and Schaeffer, 2019):

$$\ln(Wage_{ijt}) = \beta_0 + u_{0jt} + (\beta_1 + u_{1jt})Union_{ijt} + \beta_2 SC_j + \beta_3 (Union_{ijt} \times SC_j) + \beta_4 X_{ijt} + \beta_5 G_{jt} + e_{ijt} \quad (2)$$

where, in addition to the same notations in Equation 1, u_{1jt} represent the normally distributed random slopes for union membership. The parameter of interest is the vector β_3 , which, under the identification assumptions, captures the effect of counties' social capital levels on the union wage effect in Estimand 2.

In all regression models, I standardise the social capital measures across all counties in the final pooled dataset for better interpretation and comparisons. I include the quadratic term to account for the non-linear effect of age. To facilitate model convergence, all continuous covariates are standardised. All models handle missing values using case-wise deletion.

6.2 Potential Endogeneity

Despite theoretically motivated efforts to adjust for confounding bias, the observational and cross-sectional nature of the data inevitably limits the ability to address endogeneity and thus affects the precision of the estimates.

First, the adjustment specification may be insufficient to account for all confounders, particularly unobserved ones. Although random effect models could partly account for unobserved residual cross-county-year variations, they still require the unconfoundedness

assumption to identify the causal effects (Gelman and Hill, 2006). While the assumption cannot be directly tested, the main findings remain substantively similar across alternative model specifications and subsample analyses, suggesting that the results are not driven by a narrow set of modelling assumptions.

Second, the cross-sectional design limits the analysis of the dynamic processes of union organisation and bargaining and their relationships with social contexts. In particular, the estimates cannot rule out reverse causality: for example, trade union organisation and activities may foster social connections and solidarity. Although the available social capital data do not allow us to directly inspect the relationship between union-related ties and organisation and measures of social capital, two considerations imply that such reverse causality, even if it existed, is unlikely to significantly bias and invalidate the observed associations.

First, existing empirical studies have provided suggestive evidence indicating that the reverse causal relationship may be weak. For example, Cornwell and Harrison (2004) find that trade unions in the United States are isolated from other civil organisations, suggesting limited overlap between union membership and broader associational networks. Furthermore, Saundry, Stuart and Antcliff (2012), based on a case study of British freelance workers, do not find evidence that within-union social relationships generate broader social capital.

Second, even if trade unions do produce social capital at the local level, their aggregate impact at the county level is likely constrained by scale. The social capital measures are summaries of county-level patterns of social relationships and civic associations, based on friendship networks comprising, on average, more than twenty thousand Facebook users per county (Chetty et al., 2022). Given that the current overall union density in the United States is only about 10%, the significance of trade unions in shaping county-level patterns of social relationships is likely limited. In particular, with the decentralised and workplace-based union system in the United States, any union-generated social ties are likely to be concentrated within specific workplaces or sectors. It is therefore unlikely that such localised effects would be large enough to substantially alter county-level network configurations or

the aggregate measures of social capital examined here.

To summarise, the results presented below should not be interpreted as precise estimates of the causal relationships due to possible endogeneity bias. Nonetheless, given the considerations above, the magnitude of potential bias is unlikely to be sufficient to fully account for the main patterns documented below. Overall, the empirical results should be viewed as theoretically motivated approximations of the target estimands that inform the relationship between social structural features and labour market relationships (Spirling and Stewart, 2025).

6.3 Heterogeneity and Robustness Tests

Existing research has documented substantial heterogeneity in union participation and the union wage effect across demographic groups, education levels, and sectors (i.e., public versus private) (Card, Lemieux and Riddell, 2020; Farber et al., 2021; Jäger, Naidu and Schoefer, 2024). In the United States, while unionisation was historically concentrated among low-skilled male workers in the private sector, recent evidence has shown that union participation has become increasingly strong among more educated, female and public-sector workers (Card, Lemieux and Riddell, 2020). Furthermore, given persistent racial, ethnic and gender wage inequalities in the United States labour market that favour the non-Hispanic white majority and males, existing studies have found that unions could act as an equalising institution by delivering stronger welfare improvements for disadvantaged groups (e.g. Kerrissey and Meyers, 2022; VanHeuvelen, Himmelstern and Baker, 2025).

With such heterogeneity, one may expect the effect of social capital on union participation and wage effects to also vary among different groups, if, for example, the mobilisation mechanisms may be particularly effective among certain demographic groups. Therefore, for each estimand, I estimate the models separately by four demographic groups - non-Hispanic white males, non-Hispanic white females, other males and other females - to capture the intersectional heterogeneity across race/ethnicity and gender. Although substantial hetero-

geneity exists within the Hispanic and non-white population, the current analysis focuses on the difference between the non-Hispanic white majority and other minority groups to capture the main axis of racial-ethnic stratification in the US. I also explore the heterogeneity between private and public sector workers, and workers with and without a college degree.

I also test for the temporal and contextual robustness of the results. First, counties, as administrative units, may not perfectly align with the local labour market context. Therefore, to test the results' robustness to this potential misalignment, I aggregate the group-level data to the commuting zones (CZs), which are designed to reflect the local labour markets, and re-estimate the models using the aggregated data. Second, to test the results' robustness to temporal and political changes, I estimate the models separately for the periods 2013-2016 and 2017-2019, corresponding to two presidential terms.

For simplicity, I consider only the overall main effects of the three social capital measures in the heterogeneity and robustness tests.

7 Results

7.1 Descriptive Statistics

The composition of union members has changed substantially in the past decades. Card, Lemieux and Riddell (2020) estimate that approximately half of union members in the United States and Canada are employed in the public sector, and about half are female. Descriptive statistics in Table 1 are broadly consistent with these findings. The demographic composition of members and non-members is overall similar, and about half of the union members are in the public sector, compared with about 15% of workers in the overall sample. One notable pattern concerns the educational composition: workers with postgraduate degrees are over-represented among the union members, while workers with below high school education are under-represented.

The crude difference in average hourly wages indicates that union members earn, on

	Non-member <i>N</i> = 169,698 (86.3%)	Member <i>N</i> = 27,030 (13.7%)
Age		
Mean (SD)	40.725 (12.009)	43.694 (11.296)
Range	20.000 – 65.000	20.000 – 65.000
Demographics		
Non-Hisp. White Female	25.6%	23.8%
Non-Hisp. White Male	32.3%	33.6%
Other Female	18.7%	19.2%
Other Male	23.5%	23.4%
Never Married?		
No	69.1%	74.9%
Yes	30.9%	25.1%
Highest Education		
Below High School	7.0%	4.3%
High School	22.9%	23.4%
Some College	16.0%	16.8%
Associate Degree	10.2%	11.3%
Degree	28.2%	23.2%
Postgraduate	15.6%	21.0%
Sector		
Private	90.2%	48.6%
Public	9.8%	51.4%
Logged Hourly Wages		
Mean (SD)	3.124 (0.564)	3.298 (0.455)
Range	1.837 – 4.292	1.837 – 4.292
Economic Connectedness		
Mean (SD)	-0.076 (0.934)	-0.074 (0.875)
Range	-2.881 – 3.109	-2.881 – 3.109
Network Clustering		
Mean (SD)	-0.561 (0.947)	-0.513 (1.037)
Range	-2.425 – 2.908	-2.425 – 2.908
Civic Index		
Mean (SD)	-0.331 (0.881)	-0.362 (0.917)
Range	-2.428 – 3.819	-2.428 – 3.819
State with RTW law		
No	59.3%	78.2%
Yes	40.7%	21.8%
Total Pop. (million)		
Mean (SD)	1.632 (2.372)	1.796 (2.581)
Range	0.096 – 10.095	0.096 – 10.095
Logged GDP (thousand)		
N-Miss	1497	88
Mean (SD)	17.697 (1.320)	17.768 (1.302)
Range	14.743 – 20.495	14.743 – 20.495
Republican Vote Share		
Mean (SD)	0.408 (0.153)	0.373 (0.149)
Range	0.041 – 0.883	0.041 – 0.883

Table 1: Descriptive statistics by union membership

average, 18.5% ($e^{0.17} = 1.185$) more than non-members. Because the union wage effect varies across demographic groups and sectors, Figure 1 further shows the distributions and medians of logged hourly wages by union memberships, by demographics and sectors. Panel 1a shows that the crude union wage premium is the lowest among non-Hispanic white men. For non-Hispanic white women, it is slightly larger. The difference is the largest among the Hispanic and other non-white workers, and the magnitudes are similar for males and females. Panel 1b shows that the crude wage premium is substantially greater for workers without a degree, while that for workers with a degree is slightly negative. Panel 1c shows that the difference is greater among private sector workers.

These variations appear to be driven largely by differences in the wage distribution of non-members, and the wage distributions for union members are relatively similar across sub-groups, consistent with the view that unions contribute to wage coordination. These descriptive patterns are consistent with Farber et al. (2021), which document larger union wage premiums for non-white and non-college-educated workers.

Figure 2 shows the county-level distributions of, and correlations among, the three social capital variables. All three variables are standardised. The size of the dots in the scatter plots is proportional to the number of observations in the county. The civic index exhibits a moderately strong correlation with EC and a moderate correlation with network clustering. However, there is no significant correlation between EC and clustering. Overall, the absence of strong correlations suggests that the three measures capture distinct aspects of local social context and provide sufficient independent variation for analysis.

It should be noted that, due to sample selection—particularly the restriction to observations with valid county identifiers—the county-level distributions and correlations of social capital differ somewhat from those reported in Chetty et al. (2022). In particular, the distribution of network clustering is highly concentrated in counties with small populations that were excluded from the analytical sample due to identifier unavailability. However, as discussed above, this selection should not undermine the inferential validity of the results.

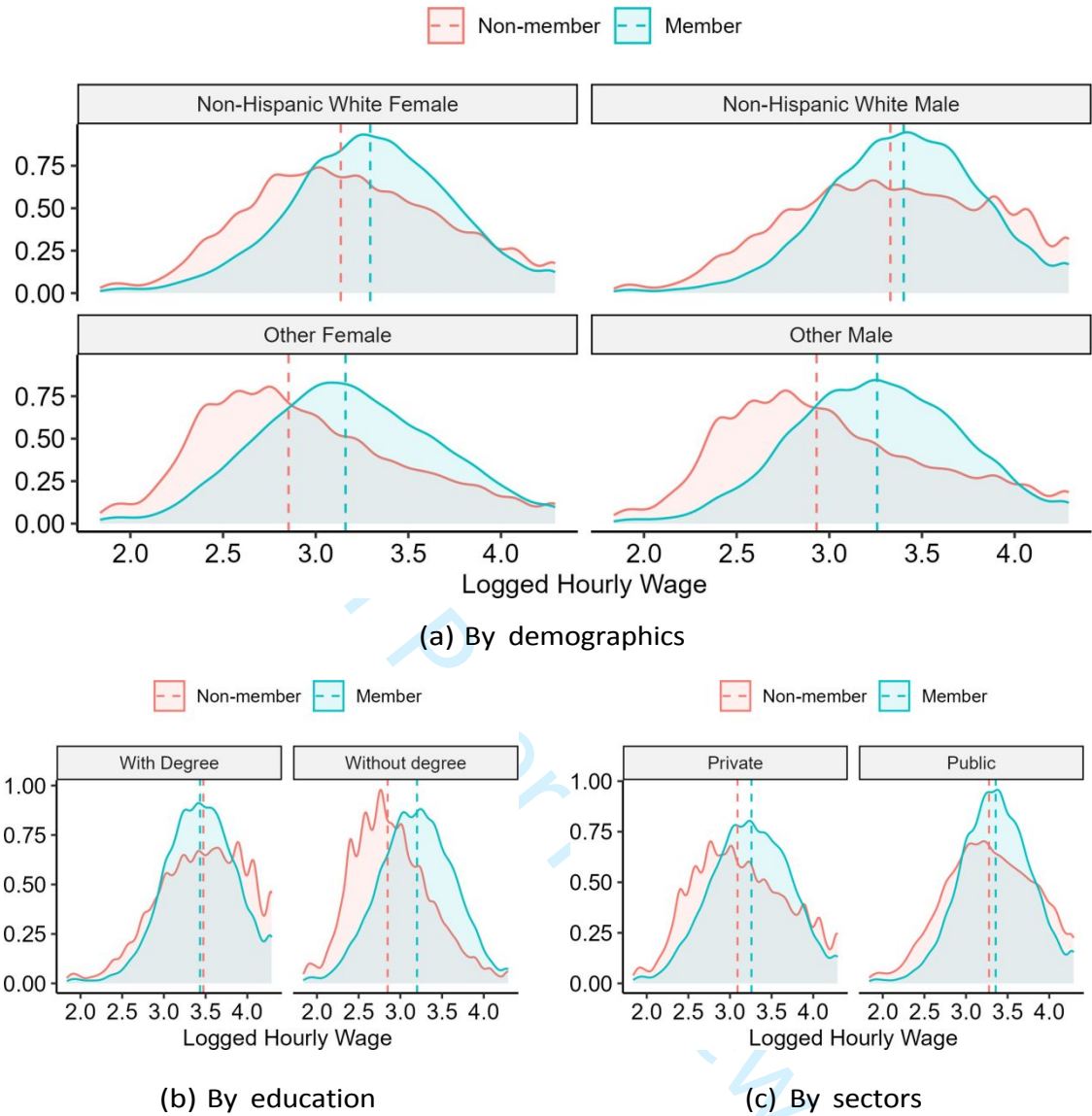
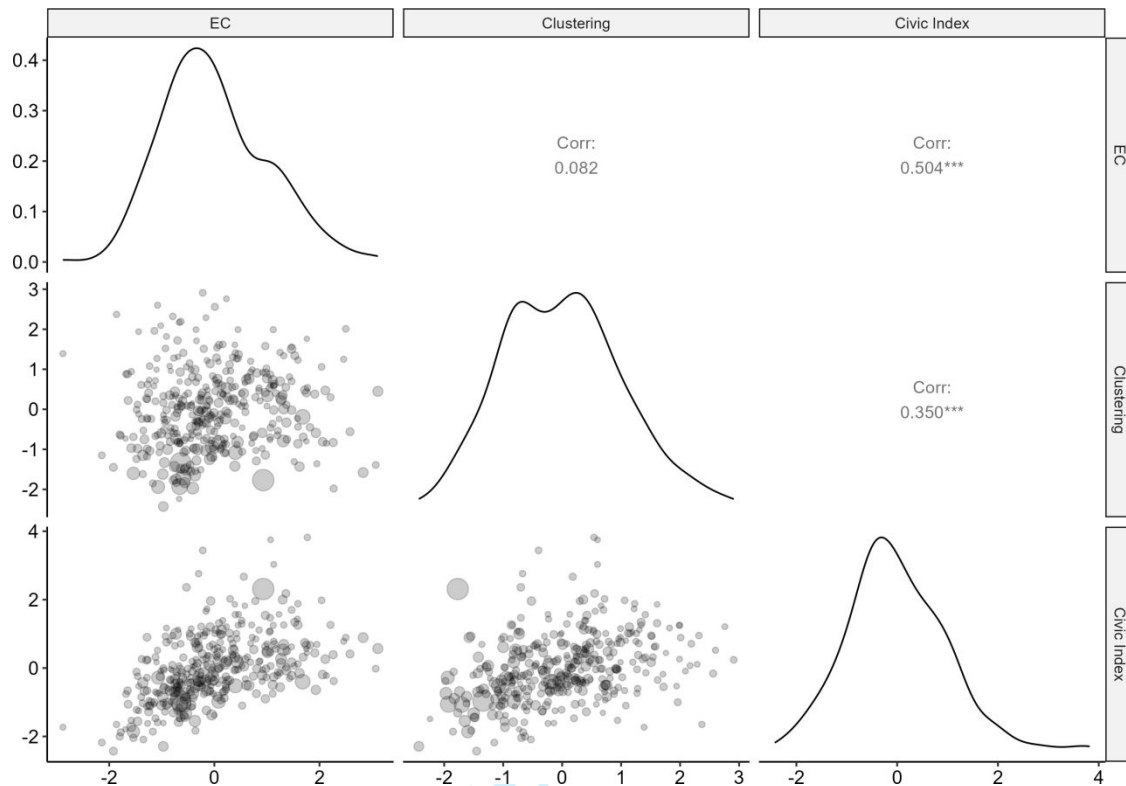


Figure 1: Distributions and medians of logged hourly wages by union membership

7.2 Union Participation

I first examine the social capital's effect on individuals' propensity for union participation. Table 2 reports model estimates for Estimand 1. Models 1.1 to 1.3 include the three social capital variables without interactions and progressively add additional controls.

Model 1.1 adjusts for individual characteristics. In Model 1.1, counties' EC and network clustering are positively associated with individuals' probability of union membership. A one standard deviation (sd) increase is on average associated with increases of 0.9 and 1.4



Note: $N = 367$. All three measures are standardised. Point sizes in the scatter plots are proportional to the number of observations in the sample.

Figure 2: County-level distributions and correlations of the three social capital variables.

percentage points (ppt), respectively ($p < 0.001$). No statistically significant association is observed for civic engagement. Model 1.2 further adjusts for group-level demographic and macroeconomic conditions. Compared to Model 1.1, the estimated coefficient for EC increases by 0.3 ppt, and that for network clustering increases by 1.7 ppt. Furthermore, the civic engagement index is now positively associated with union participation ($p < 0.001$). However, after adjusting for counties' political contexts in Model 1.3, the estimated coefficients for EC and the civic engagement index decrease substantially and become statistically insignificant at the 1% confidence level. The coefficient for network clustering also decreases but remains statistically significant and substantively meaningful - a 1 sd increase is on average associated with a 2.1 ppt increase in the probability of union participation ($p < 0.001$).

The differences between Models 1.2 and 1.3 may be due to various causal pathways. For example, the political context may be a confounder - as shown in Figure B1 - that positively

biases the causal relationship, in which case the estimates in Model 1.3 are more accurately identified. It is, however, also plausible that the political context is a mediator or collider between social capital and union participation, in which case adjusting for it negatively biases the model estimates. In practice, the true causal structure cannot be directly verified, making it difficult to adjudicate between these specifications. However, following Ding and Miratrix (2015)'s conclusion that confounding bias is generally greater than collider bias, I therefore focus on the more fully controlled specification (Model 1.3) as the preferred model.

Overall, the models indicate a robust positive conditional association between counties' network clustering and workers' probability of union participation, consistent with Hypothesis 2. EC and the civic engagement index are also estimated to have positive conditional associations with workers' union participation, though the associations are not robust under the preferred model adjustments, providing only suggestive evidence for Hypotheses 1 and 3.

Models 1.1 to 1.3 estimate the independent average effect of each type of counties' social capital on workers' union participation. However, these models do not capture potential joint effects among the three dimensions of social capital. To test such joint effects, Model 1.4 adds to Model 1.3 the full interactions among the three social capital variables. All interaction terms are statistically significant at the 5% level, indicating substantive joint effect heterogeneity for all three variables.

As stated in Hypothesis 4, the effect of network clustering on union collective action is expected to be greater in the presence of greater civic engagement. The estimated coefficient for the two-way interaction between network clustering and civic index in Model 1.4 supports the hypothesis. The effect of network clustering on union participation increases with higher levels of civic engagement. When EC is held at its mean, a one standard deviation increase in the civic index amplifies the effect of network clustering by approximately 1.8 percentage points ($p < 0.001$).

Overall, the results suggest a robust positive association between county-level network

	Model 1.1	Model 1.2	Model 1.3	Model 1.4
EC	0.009*** (0.002)	0.012*** (0.002)	0.003 (0.002)	0.002 (0.002)
Clustering	0.014*** (0.002)	0.031*** (0.002)	0.021*** (0.002)	0.023*** (0.002)
Civic Index	-0.002 (0.002)	0.009*** (0.002)	0.004* (0.002)	0.008*** (0.002)
EC:Clustering				-0.004* (0.002)
Clustering:Civic Index				0.018*** (0.002)
EC:Civic Index				-0.012*** (0.002)
EC:Clustering:Civic Index				-0.006*** (0.002)
Individual-level Ctrl.s.	Yes	Yes	Yes	Yes
Demog./Macroecon.	No	Yes	Yes	Yes
Politics	No	No	Yes	Yes
AIC	93075.907	80283.554	79610.823	79512.649
BIC	93310.268	81271.266	80618.900	80561.457
Log Likelihood	-46514.954	-40044.777	-39706.411	-39653.325
Num. obs.	196728	195360	195360	195360
Num. groups: County-Year	2068	2053	2053	2053
Var: random intercepts	0.006	0.005	0.003	0.002
Var: Residual	0.092	0.087	0.087	0.087

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Note: Individual-level controls include sex, race, ethnicity, education, marital status, age and age squared, industry, and occupation. Group-level demographic and macroeconomic controls include logged population, logged proportion of white population, logged GDP, logged unemployment rate, and logged government sector GDP contribution. Political controls include state-level RTW law status and the vote share of the Republican party in the last presidential election.

Table 2: Multilevel linear regression estimates of social capital's effect on union participation

clustering and union participation. It also suggests a positive interaction effect between a county's network clustering and the civic engagement index on workers' union participation. These findings are consistent with the hypothesised positive role of network clustering (Hypothesis 2) and with the moderating effect proposed in Hypothesis 4. The evidence for economic connectedness and civic engagement, however, remains more tentative and model-dependent.

7.3 Union Wage Effect

I next examine the effect of social capital on unions' collective bargaining power, measured by the union wage effect. First, Models 2.1 - 2.3 in Table 3 estimate the overall union wage effect. The estimates are similar across specifications with additional controls. The average union wage effect is approximately 15% ($p < 0.001$), consistent with the CPS-based estimates between 10% and 20% by Farber et al. (2021).

Models 2.4 and 2.5 add the two-way and full interactions between union membership and the social capital variables. In Model 2.4, the union wage effect increases with the level of network clustering: on average, a one standard deviation increase in clustering is associated with a 2% larger union wage premium ($p < 0.001$). This finding is consistent with Hypothesis 2. In contrast to the predictions of Hypotheses 1 and 3, the union wage effect is negatively associated with the level of EC and the civic engagement index - on average, a 1 sd increase is associated with a 1.1% ($p < 0.01$) and 2.6% ($p < 0.001$) decrease in the union wage effect, respectively. Figure 3 shows the predicted conditional union wage effect on logged wage, by each social capital variable.

Does the moderating effect of each type of social capital vary by the levels of other social capital? Estimates in Model 2.5 show that the further interaction terms are all insignificant, and the coefficients for the two-way interaction terms are similar to Model 2.4. The estimates, therefore, do not provide support for Hypothesis 4 in this specification.

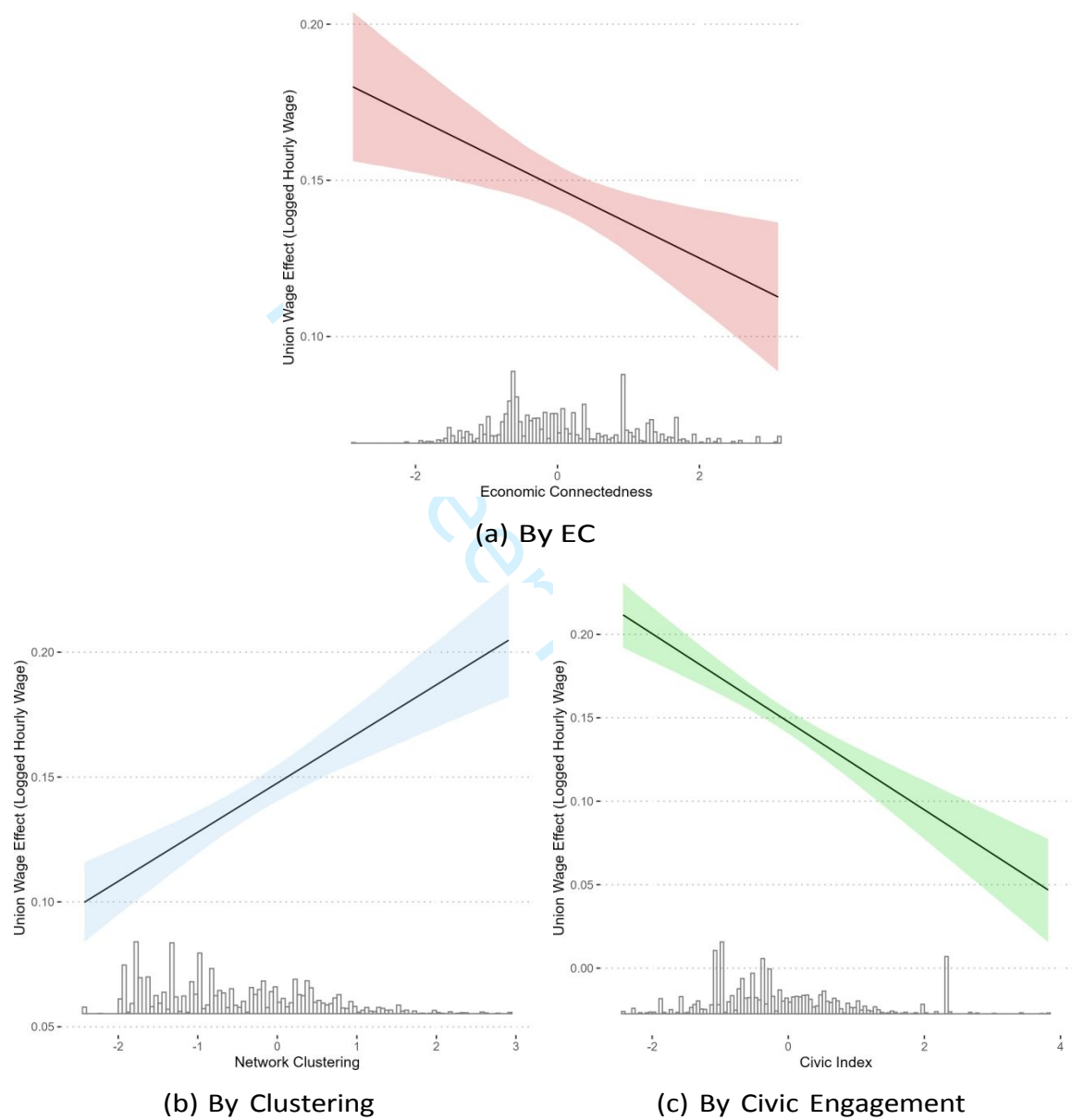
How substantively significant are these effects? VanHeuvelen (2023) estimates that the

Table 3: Multilevel estimates of the union wage effect and moderations by social capitals

	Model 2.1	Model 2.2	Model 2.3	Model 2.4	Model 2.5
Union Member	0.148*** (0.004)	0.146*** (0.004)	0.144*** (0.004)	0.148*** (0.004)	0.144*** (0.004)
Interaction with:					
EC				-0.011** (0.004)	-0.010* (0.004)
Clustering				0.020*** (0.003)	0.023*** (0.004)
Civic Index				-0.026*** (0.004)	-0.023*** (0.005)
EC:Clustering					0.001 (0.004)
Clustering:Civic Index					0.005 (0.004)
EC:Civic Index					0.004 (0.004)
EC:Clustering:Civic Index					-0.004 (0.004)
Ind. Ctrls.	Yes	Yes	Yes	Yes	Yes
Demog./Macroecon./SC	No	Yes	Yes	Yes	Yes
Politics	No	No	Yes	Yes	Yes
AIC	181769.232	179971.667	179826.265	179776.803	179847.793
BIC	182737.242	180989.926	180864.890	180845.975	180998.426
Log Likelihood	-90789.616	-89885.833	-89811.132	-89783.401	-89810.896
Num. obs.	196728	195360	195360	195360	195360
Num. groups: county_year	2068	2053	2053	2053	2053
Var: random intercepts	0.010	0.007	0.006	0.006	0.006
Var: random slopes	0.004	0.004	0.004	0.003	0.003
Cov: intercepts and slopes	-0.001	-0.001	-0.001	-0.001	-0.001
Var: Residual	0.144	0.144	0.144	0.144	0.144

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Note: Group-level demographic and macroeconomic controls include logged population, logged proportion of white population, logged GDP, logged unemployment rate, and logged government sector GDP contribution. Political controls include state-level RTW law status and the vote share of the Republican party in the last presidential election.



Note: The conditional effects are predicted from Model 2.3. The shaded area represents the 95% confidence interval.

Figure 3: Conditional union wage effect by the level of three measures of social capital

RTW law, on average, reduces the union wage effect by about 5%. The average effect of a one standard deviation change in network clustering or the civic engagement index is therefore roughly half the magnitude of the estimated RTW effect. Relative to a major institutional change such as RTW legislation, these contextual effects are substantively meaningful.

Overall, the regression result suggests a positive moderating effect of the county's network clustering on the union wage effect, thus supporting the positive effect of network clustering on union collective action (Hypothesis 2). However, contradicting Hypotheses 1 and 3, there is also evidence for the negative moderating effects of EC and the civic engagement index. While the present study cannot provide a definitive explanation, one possible, speculative interpretation is the substitution effect between union-based organisations and other forms of civic engagement. The substitution effect arises when unions and other forms of civic engagement function as substitution for each other. In that case, if union-based organisation is limited or weakened in a county, people may turn to other forms of collective engagement and support, thus creating a negative association. A similar mechanism is suggested by Diemer (2024), who finds that in the United States, in areas where traditional forms of social associations have been undermined by deindustrialisation, residents exhibit stronger solidarity and participate more in community services as a response. This interpretation, however, remains speculative and requires further investigation. Finally, there is no significant positive interaction effect of network clustering and the civic engagement index as predicted by Hypothesis 4, although the absence of a significant interaction may reflect limited statistical power.

Taken together, the results from the analysis of union participation and union wage effect show consistent evidence for the positive association between county-level network clustering and both union participation and the union wage effect, consistent with Hypothesis 2. The analysis of union participation also suggests a positive interaction effect between network clustering and civic engagement on union collective action, though the effect is not found in the analysis of the union wage effect. There is, therefore, only partial support for

Hypothesis 4. The effects of EC and the civic engagement index are more complicated. For union participation, there is no significant effect for EC and a weakly positive effect for the civic engagement index, while for the union wage effect, both EC and the civic engagement index have significant negative effects. The divergent patterns may reflect differences in the underlying processes generating union participation and wage bargaining outcomes, though more thorough explanations are beyond the scope of the present study. Overall, the evidence regarding the union wage effect does not provide consistent support for Hypotheses 1 and 3.

7.4 Heterogeneity and Robustness Tests

How do the effects of social capital on union participation and union wage effects vary by individual characteristics? And how robust are the results to temporal and spatial factors? This section explores these questions and reports the results of heterogeneity and robustness tests.

7.4.1 Union Participation

Figure 4 reports the results of the heterogeneity and robustness tests for social capital's effect on union participation. The point estimates and 95% confidence intervals are obtained by estimating Model 1.3 separately for each sub-sample.

Panel 4a shows that while the effects of EC and the civic engagement index are similar across demographic groups and statistically insignificant, the positive effect of network clustering is larger for non-Hispanic white workers by approximately 0.1 percentage points (ppt), nearly doubling the magnitude observed for other workers. There is no significant heterogeneity by sex. This pattern may suggest that the network-based mobilisation mechanisms operate differently across demographic contexts, potentially reflecting variation in local social organisation and associational structures. For example, Sampson (2013) documents that white-majority neighbourhoods in Chicago have higher levels of collective efficacy than black-majority ones, which leads to weaker social control and collective action in black

neighbourhoods. The observed heterogeneity may therefore reflect differences in broader social environments that are not fully captured by the civic index.

Panel 4b shows the heterogeneity by the level of education. It shows that there is no significant difference in the effect of EC and network clustering on workers' union participation between workers with and without a degree. The effect of the civic engagement index is, however, stronger and significantly positive among workers with a degree, potentially reflecting differences in engagement with collective norms.

Finally, Panel 4c shows substantive heterogeneity in the effect of network clustering and the civic engagement index by sectors. It shows that while the effect of network clustering is significantly positive for both private and public sector workers, the average effect for public sector workers (7.8 ppt) is more than eight times larger than that for private sector workers (0.9 ppt). Furthermore, the effect of the civic engagement index is estimated to be almost 0 for private sector workers but significantly positive for public sector workers. These differences suggest meaningful heterogeneity in the mechanisms and institutional space for union mobilisation across sectors.

Panels 4d and 4e show the results of the robustness tests. Panel 4d presents estimates obtained separately for the two presidential periods (2013–2016 and 2017–2019). It shows that, with the exception that the effect of EC appears larger in 2017–2019, the estimates are overall consistent in the two periods. Panel 4e shows the model estimates when fitted with county-level and CZ-aggregated data. Estimates based on county-level and CZ-aggregated data are broadly similar for EC and network clustering, while the estimated effect of the civic engagement index is larger under CZ aggregation. Overall, while the quantitative differences may reflect cross-time differences or labour dynamics at different geographic levels, the main patterns are qualitatively robust.

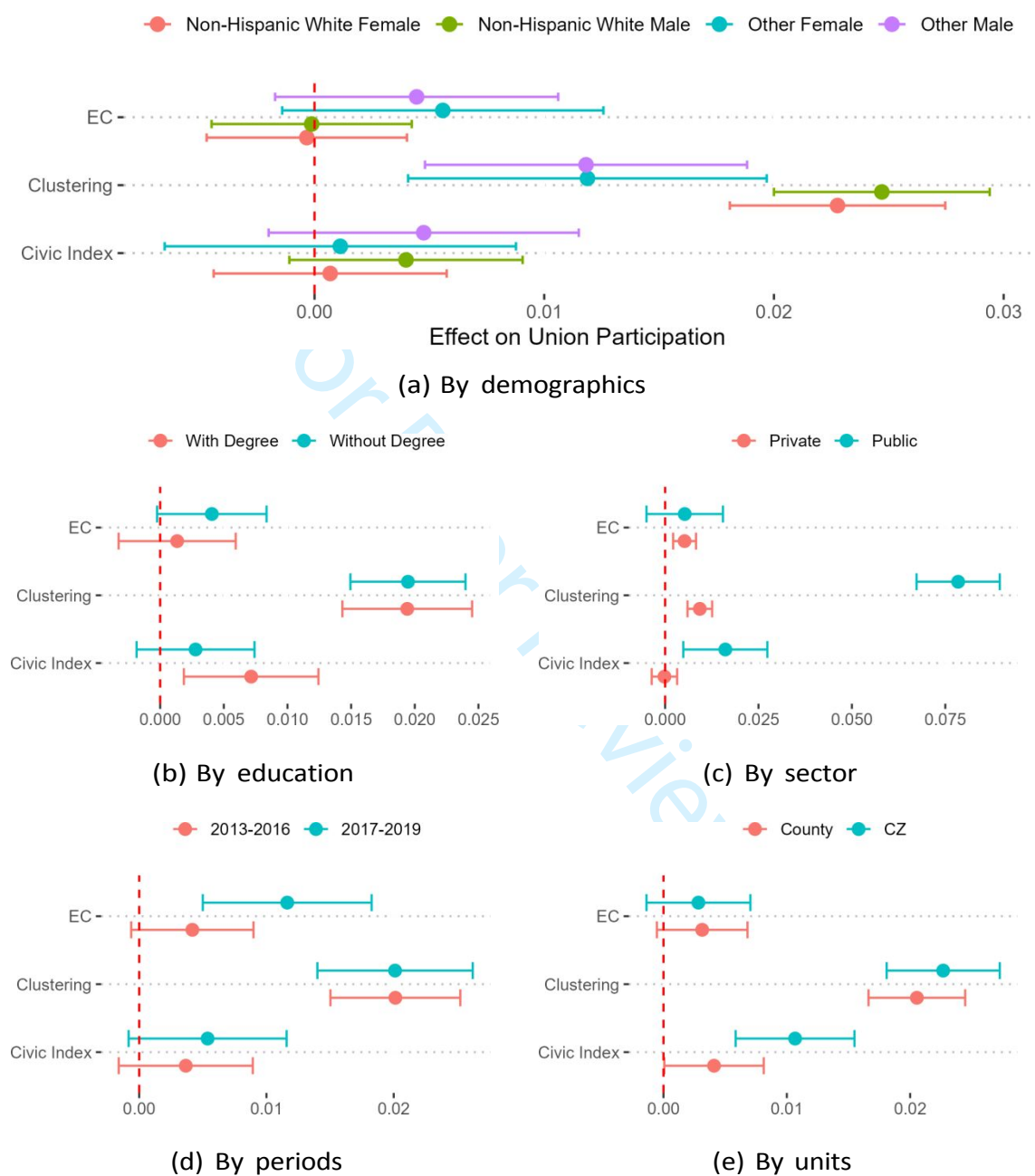


Figure 4: Heterogeneity and robustness tests for social capital’s effect on union participation

7.4.2 Union Wage Effect

Figure 5 reports the distributions of group-specific proportional union wage effects. The estimates are obtained by fitting Model 2.3 separately for each sub-sample.

Panel 5a indicates substantial heterogeneity in the distribution of the union wage effect across demographic groups. In terms of central tendency, the average union wage effect (i.e., the fixed-effect estimate) is largest for non-white men, followed by white men and non-white women with similar magnitudes, and smallest for white women. The average effect for non-white men is more than twice that for white women. However, in terms of dispersion, the distribution for non-white men is the most dispersed, indicating greater effect heterogeneity. Panel 5b shows that the average union wage effect is significantly larger for workers without a college degree, approximately twice that for workers with a college degree. Overall, these patterns are consistent with Farber et al. (2021)'s result that the union wage premium is greater for non-white workers and workers without a college degree. In addition to these results, Panel 5a also shows that within each racial and ethnic group, the union wage effect is smaller for female workers. Finally, Panel 5c shows that the average union wage effect for private sector workers is more than twice that for public sector workers, and dispersion is also greater in the private sector.

In terms of the cross-year and cross-level robustness of the result, the effect distributions in Panels 5d and 5e largely overlap, suggesting that the overall patterns are robust across temporal and spatial specifications.

Figure 6 reports the results of the heterogeneity and robustness tests for social capital's moderating effect on the union wage effect. Panel 6a shows that the negative moderating effect of EC on the union wage premium is statistically significant only for non-white workers at the 95% confidence level. The moderating effect of network clustering is more heterogeneous by sex, with the positive moderating effect appearing larger for female workers. For the civic engagement index, no consistent pattern emerges, though the negative moderating effect is the weakest and insignificant for non-white females.

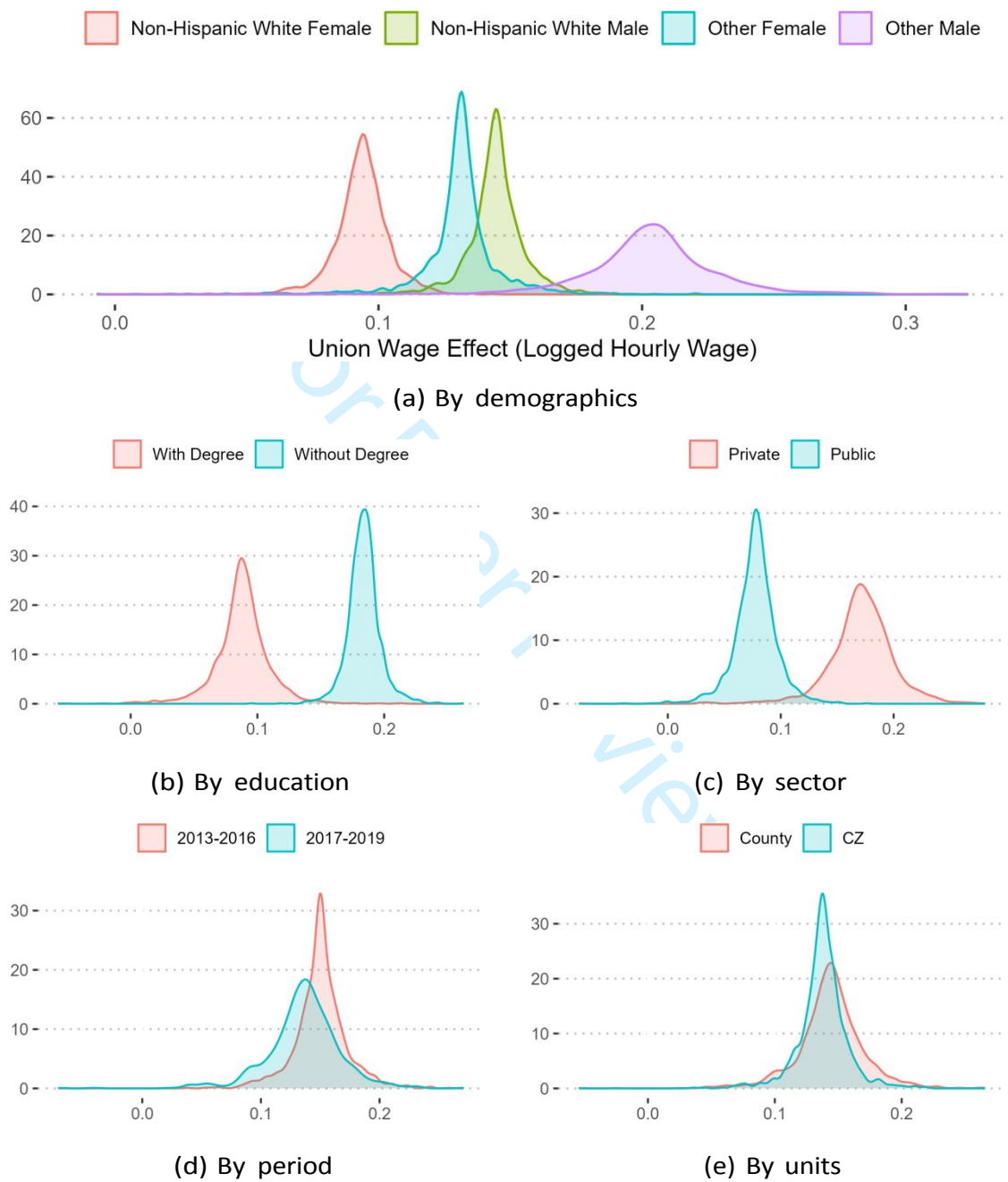
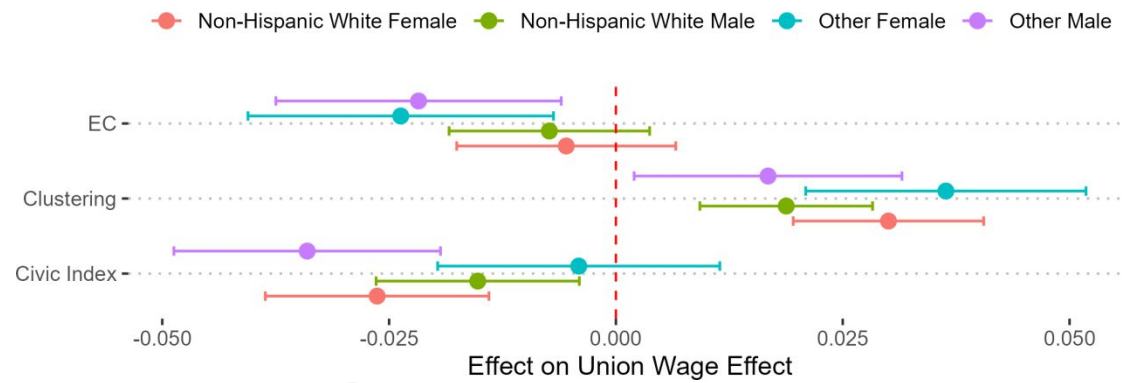
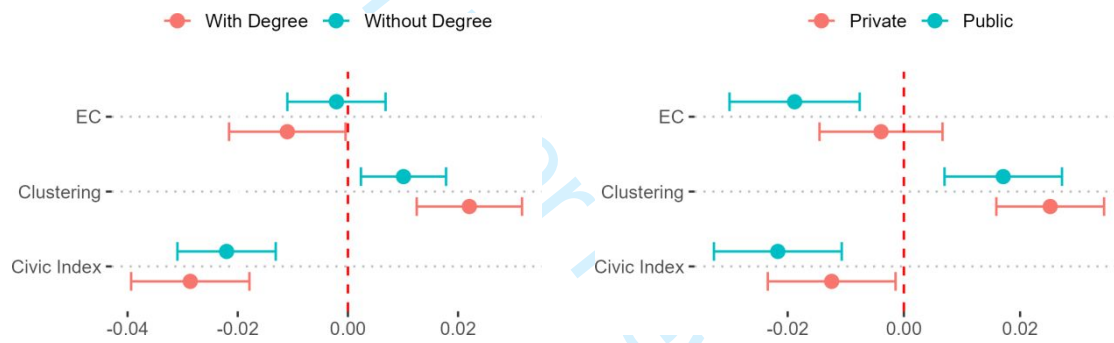


Figure 5: Distribution of group-specific union wage effect

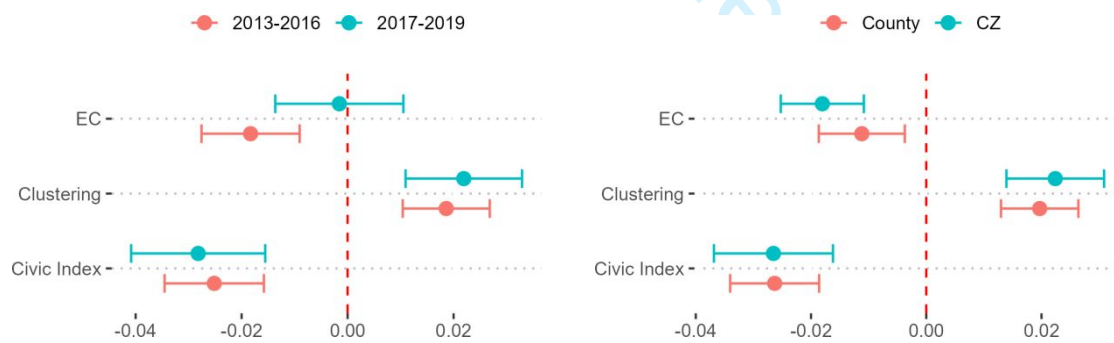


(a) By demographics



(b) By education

(c) By sector



(d) By period

(e) By units

Figure 6: Heterogeneity and robustness tests for social capital's moderating effect on union wage effect

Panel 6b shows that the absolute magnitudes of the moderating effects of all three social capital measures are larger for workers with a college degree. Finally, Panel 6c shows that the negative moderating effects of EC and the civic engagement index are more pronounced for public sector workers, while the positive moderating effect of network clustering is comparatively weaker. Panels 6d and 6e show that the results are mostly consistent across the temporal and spatial contexts, except that the moderating effect of EC is close to zero in the 2017–2019 period.

Taken together, what do the heterogeneity and robustness tests reveal? First, in terms of effect heterogeneity, the results indicate meaningful heterogeneity across multiple dimensions, with different patterns for union participation and the union wage effect. Although the underlying mechanisms driving these subgroup differences require further investigation, they provide suggestive evidence for substantive differences in the processes of unionisation and collective wage setting. Second, despite the variations, the association between network clustering and union outcomes remains positive and statistically significant across subgroups and robustness specifications. These findings provide additional support for the positive role of network clustering in enabling union collective action.

8 Discussion and Conclusion

At its core, trade unions involve a problem of collective action. Whether workers can organise, mobilise, and bargain effectively depends not only on market conditions or institutional rules, but also on the social structure within which coordination becomes possible. This study, therefore, examines the social foundations of trade union collective action by focusing on the enabling role of civil society and local social structure. Drawing on theories of collective action, social capital and social networks, I identify three types of social capital: economic connectedness, network clustering and civic engagement that may enable union collective action through resource-based, network and cultural mechanisms.

Using novel county-level measures of social capital linked to individual labour market outcomes in the United States, multilevel model estimates show that higher levels of network clustering are consistently associated with a higher probability of union participation and a stronger union wage effect. In contrast, evidence for the effect of economic connectedness and civic engagement is weaker and more inconsistent. The findings clarify the micro-social foundations of union collective action. In doing so, the study highlights the importance of local social contexts, particularly network structure, for understanding labour market relationships and outcomes alongside market and institutional factors. Ultimately, the study bridges and contributes to the literature on civil society, collective action and labour market relationships by presenting a more comprehensive framework for analysing the interactions of social, market and institutional factors in shaping labour market relationships and outcomes.

In practical terms, the study highlights the importance of local social structure, alongside market and institutional forces, in enabling worker organisation and shaping labour market outcomes. In contexts such as the United States, where decentralised labour market institutions and increasing firm market power have constrained workers' capacity for collective bargaining, the findings point to a potential "silver lining" for labour organisations. Despite these constraints, workers may sustain and develop collective power through locally embedded social connections and autonomous forms of association. This implies that labour power may, to some extent, be rebuilt from the bottom up through strengthening social ties and community-based organisation, rather than relying solely on institutional reform.

Nonetheless, the findings should be interpreted in light of methodological limitations. First, the analysis relies on observational, cross-sectional data, and despite extensive methodological considerations, causal interpretation remains limited by potential unobserved confounding and reverse causality. Future research using longitudinal or quasi-experimental designs would help to more precisely identify the causal effects of social capital on union outcomes. Second, the Facebook-based social capital measures are likely imperfect proxies for the structural features that affect collective action. In particular, the ability to infer

the within-union mobilisation processes is limited by the absence of union-specific network information. More fine-grained network data would allow more direct examinations of the collective action mechanisms.

The study opens new directions for future research to examine the embeddedness of labour organisations in civil society and the interaction of market, social and institutional factors in shaping labour market relationships and outcomes. For example, the divergent effects across different dimensions of social capital, particularly the weak or negative associations for economic connectedness and civic engagement, raise important questions about the mechanisms and conditions under which different aspects of civil society support or substitute for union organisation, which require further theoretical formulation and empirical evidence. Extending the theoretical framework, future research could also examine other mechanisms by which civil society strengthens trade unions, such as building resilience to institutional shocks like the passage of the RTW law (VanHeuvelen, 2020, 2023). Finally, with the rise of alternative, non-union forms of labour activism, future research could examine the changing role of civil society, particularly digital networks, in enabling such movements and shaping labour market outcomes (Rosenfeld, 2019).

References

- Abbott, Andrew. 1997. "Seven Types of Ambiguity." *Theory and Society* 26(2/3):357–391.
- Alder, Simeon D., David Lagakos and Lee Ohanian. 2023. "Labor Market Conflict and the Decline of the Rust Belt." *Journal of Political Economy* 131(10):2780–2824.
- Barth, Erling, Alex Bryson and Harald Dale-Olsen. 2020. "Union Density Effects on Productivity and Wages." *The Economic Journal* 130(631):1898–1936.
- Bhuller, Manudeep, Karl Ove Moene, Magne Mogstad and Ola L. Vestad. 2022. "Facts and Fantasies about Wage Setting and Collective Bargaining." *Journal of Economic Perspectives* 36(4):29–52.
- Blanchflower, David G. and Alex Bryson. 2022. "Union Membership Peaks in Midlife." *British Journal of Industrial Relations* 60(1):124–151.
- Booth, Alison L. 1985. "The Free Rider Problem and a Social Custom Model of Trade Union Membership*." *The Quarterly Journal of Economics* 100(1):253–261.
- Bourdieu, Pierre. 1986. The Forms of Capital. In *Handbook of Theory and Research for the Sociology of Education*, ed. John G. Richardson. Greenwood New York, NY p. 258.
- Brady, David and Benjamin Sosnaud. 2010. The Politics of Economic Inequality. In *Handbook*

- of Politics: State and Society in Global Perspective*, ed. Kevin T. Leicht and J. Craig Jenkins. Handbooks of Sociology and Social Research New York, NY: Springer pp. 521–541.
- Brady, David and Kevin T. Leicht. 2008. "Party to Inequality: Right Party Power and Income Inequality in Affluent Western Democracies." *Research in Social Stratification and Mobility* 26(1):77–106.
- Brady, David, Regina S. Baker and Ryan Finnigan. 2013. "When Unionization Disappears: State-Level Unionization and Working Poverty in the United States." *American Sociological Review* 78(5):872–896.
- Bryson, Alex and David G. Blanchflower. 2009. Trade Union Decline and the Economics of the Workplace. In *The Evolution of the Modern Workplace*, ed. William Brown, Alex Bryson, Keith Whitfield and John Forth. United Kingdom: Cambridge University Press.
- Card, David. 2022. "Who Set Your Wage?" *American Economic Review* 112(4):1075–1090.
- Card, David, Thomas Lemieux and W. Craig Riddell. 2003. "Unionization and Wage Inequality: A Comparative Study of the U.S, the U.K., and Canada."
- Card, David, Thomas Lemieux and W. Craig Riddell. 2020. "Unions and Wage Inequality: The Roles of Gender, Skill and Public Sector Employment." *Canadian Journal of Economics/Revue canadienne d'économique* 53(1):140–173.
- Centola, Damon and Michael Macy. 2007. "Complex Contagions and the Weakness of Long Ties1." *American Journal of Sociology* .
- Chetty, Raj, Matthew O. Jackson, Theresa Kuchler, Johannes Stroebe, Nathaniel Hendren, Robert B. Fluegge, Sara Gong, Federico Gonzalez, Armelle Grondin, Matthew Jacob, Drew Johnston, Martin Koenen, Eduardo Laguna-Muggenburg, Florian Mudekereza, Tom Rutter, Nicolaj Thor, Wilbur Townsend, Ruby Zhang, Mike Bailey, Pablo Barberá, Monica Bhole and Nils Wernerfelt. 2022. "Social Capital I: Measurement and Associations with Economic Mobility." *Nature* 608(7921):108–121.
- Coleman, James S. 1988. "Social Capital in the Creation of Human Capital." *American Journal of Sociology* 94:S95–S120.
- Cornwell, Benjamin and Jill Ann Harrison. 2004. "Union Members and Voluntary Associations: Membership Overlap as a Case of Organizational Embeddedness." *American Sociological Review* 69(6):862–881.
- Diani, Mario and Ann Mische. 2015. Network Approaches and Social Movements. In *The Oxford Handbook of Social Movements*, ed. Donatella della Porta and Mario Diani. Oxford University Press p. 0.
- Diani, Mario and Doug McAdam, eds. 2003. *Social Movements and Networks: Relational Approaches to Collective Action*. 1 ed. Oxford University Press Oxford.
- Dickens, William T. and Jonathan S. Leonard. 1985. "Accounting for the Decline in Union Membership, 1950–1980." *ILR Review* 38(3):323–334.
- Diemer, Andreas. 2024. "Divided We Fall? The Effect of Manufacturing Decline on the Social Capital of US Communities." *Journal of Regional Science* 64(1):80–107.
- Ding, Peng and Luke W. Miratrix. 2015. "To Adjust or Not to Adjust? Sensitivity Analysis of M-Bias and Butterfly-Bias." *Journal of Causal Inference* 3(1):41–57.
- Dixon, Marc and Andrew W. Martin. 2012. "We Can't Win This on Our Own: Unions, Firms, and Mobilization of External Allies in Labor Disputes." *American Sociological Review* 77(6):946–969.

- Durlauf, Steven N. 2002. "On the Empirics of Social Capital." *The Economic Journal* 112(483):F459–F479.
- Ebbinghaus, Bernhard, Claudia Göbel and Sebastian Koos. 2011. "Social Capital, 'Ghent' and Workplace Contexts Matter: Comparing Union Membership in Europe." *European Journal of Industrial Relations* 17(2):107–124.
- Ebbinghaus, Bernhard and Jelle Visser. 1999. "When Institutions Matter: Union Growth and Decline in Western Europe, 1950–1995." *European Sociological Review* 15(2):135–158.
- Edwards, Bob and John D. McCarthy. 2007. Resources and Social Movement Mobilization. In *The Blackwell Companion to Social Movements*, ed. David A. Snow, Sarah A. Soule and Hanspeter Kriesi. Oxford, UK: Blackwell Publishing Ltd pp. 116–152.
- Farber, Henry S, Daniel Herbst, Ilyana Kuziemko and Suresh Naidu. 2021. "Unions and Inequality over the Twentieth Century: New Evidence from Survey Data*." *The Quarterly Journal of Economics* 136(3):1325–1385.
- Fine, Janice. 2005. "Community Unions and the Revival of the American Labor Movement." *Politics & Society* 33(1):153–199.
- Firpo, Sergio, Nicole M. Fortin and Thomas Lemieux. 2009. "Unconditional Quantile Regressions." *Econometrica* 77(3):953–973.
- Fortin, Nicole M., Thomas Lemieux and Neil Lloyd. 2021. "Labor Market Institutions and the Distribution of Wages: The Role of Spillover Effects." *Journal of Labor Economics* 39(S2):S369–S412.
- Freeman, Richard B. and James L. Medoff. 1992. *What Do Unions Do?* 5. [print.] ed. New York: Basic Books.
- Fukuyama, Francis. 1995. *Trust: The Social Virtues and the Creation of Prosperity*. Free Press.
- Fukuyama, Francis. 2001. "Social Capital, Civil Society and Development." *Third world quarterly* 22(1):7–20.
- Gelman, Andrew and Jennifer Hill. 2006. *Data Analysis Using Regression and Multi-level/Hierarchical Models*. 1 ed. Cambridge University Press.
- Gould, Roger V. 1991. "Multiple Networks and Mobilization in the Paris Commune, 1871." *American Sociological Review* 56(6):716.
- Gould, Roger V. 1993a. "Collective Action and Network Structure." *American Sociological Review* 58(2):182–196.
- Gould, Roger V. 1993b. "Trade Cohesion, Class Unity, and Urban Insurrection: Artisanal Activism in the Paris Commune." *American Journal of Sociology* 98(4):721–754.
- Granovetter, Mark. 1978. "Threshold Models of Collective Behavior." *American Journal of Sociology* 83(6):1420–1443.
- Granovetter, Mark. 2018. *Getting a Job: A Study of Contacts and Careers*. University of Chicago press.
- Hardin, Russell. 1971. "Collective Action as an Agreeable N-Prisoners' Dilemma." *Behavioral Science* 16(5):472–481.
- Heckathorn, Douglas D. 1988. "Collective Sanctions and the Creation of Prisoner's Dilemma Norms." *American Journal of Sociology* 94(3):535–562.
- Heckathorn, Douglas D. 1989. "Collective Action and the Second-Order Free-Rider Problem." *Rationality and Society* 1(1):78–100.
- Heisig, Jan Paul and Merlin Schaeffer. 2019. "Why You Should Always Include a Ran-

- dom Slope for the Lower-Level Variable Involved in a Cross-Level Interaction.” *European Sociological Review* 35(2):258–279.
- Hirsch, Barry T. and Edward J. Schumacher. 2004. “Match Bias in Wage Gap Estimates Due to Earnings Imputation.” *Journal of Labor Economics* 22(3):689–722.
- Hyman, Richard. 2007. “How Can Trade Unions Act Strategically?” *Transfer: European Review of Labour and Research* 13(2):193–210.
- Inglehart, Ronald. 1997. *Modernization and Postmodernization: Cultural, Economic, and Political Change in 43 Societies*. Princeton, NJ: Princeton Univ. Pr.
- Jackman, Robert W. and Ross A. Miller. 1998. “SOCIAL CAPITAL AND POLITICS.” *Annual Review of Political Science* 1(Volume 1, 1998):47–73.
- Jacobs, David and Lindsey Myers. 2014. “Union Strength, Neoliberalism, and Inequality: Contingent Political Analyses of U.S. Income Differences since 1950.” *American Sociological Review* 79(4):752–774.
- Jäger, Simon, Suresh Naidu and Benjamin Schoefer. 2024. “Collective Bargaining, Unions, and the Wage Structure: An International Perspective.”
- Kerrissey, Jasmine and Nathan Meyers. 2022. “Public-Sector Unions as Equalizing Institutions: Race, Gender, and Earnings.” *ILR Review* 75(5):1215–1239.
- Korpi, Walter. 1978. *The Working class in welfare capitalism: work, unions and politics in Sweden*. International library of sociology London Henley Boston: Routledge and K. Paul.
- Korpi, Walter. 1985. “Power Resources Approach vs. Action and Conflict: On Causal and Intentional Explanations in the Study of Power.” *Sociological Theory* 3(2):31–45.
- Lin, Nan. 2001. Building a Network Theory of Social Capital. In *Social Capital*, ed. Rene Dubos. Routledge.
- Lin, Nan, Walter M. Ensel and John C. Vaughn. 1981. “Social Resources and Strength of Ties: Structural Factors in Occupational Status Attainment.” *American Sociological Review* 46(4):393–405.
- Lundberg, Ian, Rebecca Johnson and Brandon M. Stewart. 2021. “What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory.” *American Sociological Review* 86(3):532–565.
- Manning, Alan. 2021. “Monopsony in Labor Markets: A Review.” *ILR Review* 74(1):3–26.
- Massenkoff, Maxim and Nathan Wilmers. 2024. “Economic Outcomes of Strikers in an Era of Weak Unions.” *Journal of Labor Economics* 42(1):25–51.
- McClurg, Scott D. 2003. “Social Networks and Political Participation: The Role of Social Interaction in Explaining Political Participation.” *Political Research Quarterly* 56(4):449–464.
- Morgan, Stephen L. and Christopher Winship. 2014. *Counterfactuals and Causal Inference: Methods and Principles for Social Research*. Analytical Methods for Social Research 2 ed. Cambridge: Cambridge University Press.
- Murphy, Richard. 2020. “Why Unions Survive: Understanding How Unions Overcome the Free-Rider Problem.” *Journal of Labor Economics* 38(4):1141–1188.
- Murray, Gregor. 2017. “Union Renewal: What Can We Learn from Three Decades of Research?” *Transfer: European Review of Labour and Research* 23(1):9–29.
- Newman, Mark E. J. 2018. *Networks*. Second edition ed. Oxford: Oxford University Press.
- Nissen, Bruce and Paul Jarley. 2005. “Unions as Social Capital: Renewal through a Return to the Logic of Mutual Aid?” *Labor Studies Journal* 29(4):1–26.

- Olson, Mancur. 2009. *The Logic of Collective Action: Public Goods and the Theory of Groups, Second Printing with a New Preface and Appendix*. Harvard University Press.
- Peluso, Alina, Joseph Tuccillo, Kevin Sparks, Anuj Kapadia and Heidi A. Hanson. 2024. "Spatial Analysis of Social Capital and Community Heterogeneity at the United States County Level." *Applied Geography* 162:103168.
- Portes, Alejandro. 1998. "Social Capital: Its Origins and Applications in Modern Sociology." *Annual Review of Sociology* 24(Volume 24, 1998):1–24.
- Putnam, Robert D. 1994. *Making Democracy Work: Civic Traditions in Modern Italy*. Princeton University Press.
- Putnam, Robert D. 2000. *Bowling Alone: The Collapse and Revival of American Community*. Simon and Schuster.
- Reuter, Marta, Filip Wijkström and Michael Meyer. 2014. Who Calls the Shots? The Real Normative Power of Civil Society. In *Modernizing Democracy*, ed. Matthias Freise and Thorsten Hallmann. New York, NY: Springer New York pp. 71–82.
- Reynolds, Megan M. and David Brady. 2012. "Bringing You More Than the Weekend: Union Membership and Self-rated Health in the United States." *Social Forces* 90(3):1023–1049.
- Rhodes, Alec P. 2026. "The Wealth Returns to a Unionized Career." *Social Forces* 104(3):1071–1095.
- Richman, Barak. 2002. "Community Enforcement of Informal Contracts: Jewish Diamond Merchants in New York." *John M. Olin Center for Law, Economics & Business, Discussion Paper No. 384*.
- Richman, Barak D. 2006. "How Community Institutions Create Economic Advantage: Jewish Diamond Merchants in New York." *Law & Social Inquiry* 31(2):383–420.
- Rosenfeld, Jake. 2019. "US Labor Studies in the Twenty-First Century: Understanding Laborism Without Labor." *Annual Review of Sociology* 45(1):449–465.
- Rupasingha, Anil, Stephan J. Goetz and David Freshwater. 2006. "The Production of Social Capital in US Counties." *The Journal of Socio-Economics* 35(1):83–101.
- Sampson, Robert J. 2013. *Great American City: Chicago and the Enduring Neighborhood Effect*. Chicago, IL: University of Chicago Press.
- Saundry, Richard, Mark Stuart and Valerie Antcliff. 2012. "Social Capital and Union Revitalization: A Study of Worker Networks in the UK Audio-Visual Industries: Social Capital and Union Revitalization." *British Journal of Industrial Relations* 50(2):263–286.
- Schaller, Zachary. 2023. "Decomposing the Decline of Unions: Revisiting Sectoral and Regional Shifts." *ILR Review* 76(2):387–411.
- Siegel, David A. 2009. "Social Networks and Collective Action." *American Journal of Political Science* 53(1):122–138.
- Skocpol, Theda. 1997. "The Tocqueville Problem: Civic Engagement in American Democracy." *Social Science History* 21(4):455–479.
- Spirling, Arthur and Brandon M. Stewart. 2025. "What Good Is a Regression? Inference to the Best Explanation and the Practice of Political Science Research." *The Journal of Politics*.
- Tapia, Maite, Christian L. Ibsen and Thomas A. Kochan. 2015. "Mapping the Frontier of Theory in Industrial Relations: The Contested Role of Worker Representation." *Socio-Economic Review* 13(1):157–184.
- U.S. Census Bureau. 2022. Improving Disclosure Avoidance Procedures for the Current

Population Survey Public Use File.

- Vâlsan, Călin, Zizi Goschin and Elena Druică. 2023. "The Measurement of Social Capital in America: A Reassessment." *Social Indicators Research* 165(1):135–161.
- VanHeuvelen, Tom. 2018. "Moral Economies or Hidden Talents? A Longitudinal Analysis of Union Decline and Wage Inequality, 1973–2015." *Social Forces* 97(2):495–530.
- VanHeuvelen, Tom. 2020. "The Right to Work, Power Resources, and Economic Inequality." *American Journal of Sociology* 125(5):1255–1302.
- VanHeuvelen, Tom. 2023. "The Right to Work and American Inequality." *American Sociological Review* p. 00031224231197630.
- VanHeuvelen, Tom and David Brady. 2022. "Labor Unions and American Poverty." *ILR Review* 75(4):891–917.
- VanHeuvelen, Tom, Jessie Himmelstern and Regina Baker. 2025. "Public Sector Unionization and Black-White Earnings Inequality." *Social Science Research* 132:103238.
- VanHeuvelen, Tom, Xiaowen Han and Jane VanHeuvelen. 2025. "The Mortality Implications of a Unionized Career." *Social Science & Medicine* 365:117620.
- Wallerstein, Michael. 1999. "Wage-Setting Institutions and Pay Inequality in Advanced Industrial Societies." *American Journal of Political Science* 43(3):649–680.
- Wallerstein, Michael and Bruce Western. 2000. "Unions in Decline? What Has Changed and Why." *Annual Review of Political Science* 3(1):355–377.
- Western, Bruce. 1993. "Postwar Unionization in Eighteen Advanced Capitalist Countries." *American Sociological Review* 58(2):266–282.
- Western, Bruce. 1994. "Institutional Mechanisms for Unionization in Sixteen OECD Countries: An Analysis of Social Survey Data*." *Social Forces* 73(2):497–519.
- Western, Bruce. 1995. "A Comparative Study of Working-Class Disorganization: Union Decline in Eighteen Advanced Capitalist Countries." *American Sociological Review* 60(2):179–201.
- Western, Bruce and Jake Rosenfeld. 2011. "Unions, Norms, and the Rise in U.S. Wage Inequality." *American Sociological Review* 76(4):513–537.
- Wilmers, Nathan. 2017. "Labor Unions as Activist Organizations: A Union Power Approach to Estimating Union Wage Effects." *Social Forces* 95(4):1451–1478.
- Yeh, Chen, Claudia Macaluso and Brad Hershbein. 2022. "Monopsony in the US Labor Market." *American Economic Review* 112(7):2099–2138.

Appendix A. CPS-MORG Descriptive Statistics

	All (N=475,397)	With County ID (N=196,857)
Age		
Mean (SD)	41.127 (12.011)	41.134 (11.957)
Range	20.000 - 65.000	20.000 - 65.000
Demographics		
Non-Hispanic White Female	133544 (28.1%)	49861 (25.3%)
Non-Hispanic White Male	170765 (35.9%)	63873 (32.4%)
Other Female	75764 (15.9%)	36900 (18.7%)
Other Male	95324 (20.1%)	46224 (23.5%)
Never Married?		
Mean (SD)	0.282 (0.450)	0.301 (0.459)
Range	0.000 - 1.000	0.000 - 1.000
Highest Education		
Below High School	30386 (6.4%)	13075 (6.6%)
High School	118145 (24.9%)	45200 (23.0%)
Some College	79547 (16.7%)	31727 (16.1%)
Associate Degree	53333 (11.2%)	20446 (10.4%)
Degree	123597 (26.0%)	54218 (27.5%)
Postgraduate	70389 (14.8%)	32191 (16.4%)
Sector		
Private	398145 (83.8%)	166262 (84.5%)
Public	77252 (16.2%)	30595 (15.5%)
Logged Hourly Wages		
Mean (SD)	3.109 (0.540)	3.148 (0.553)
Range	1.837 - 4.292	1.837 - 4.292

Table A1: Descriptive statistics for observations with valid county identifiers and all observations.

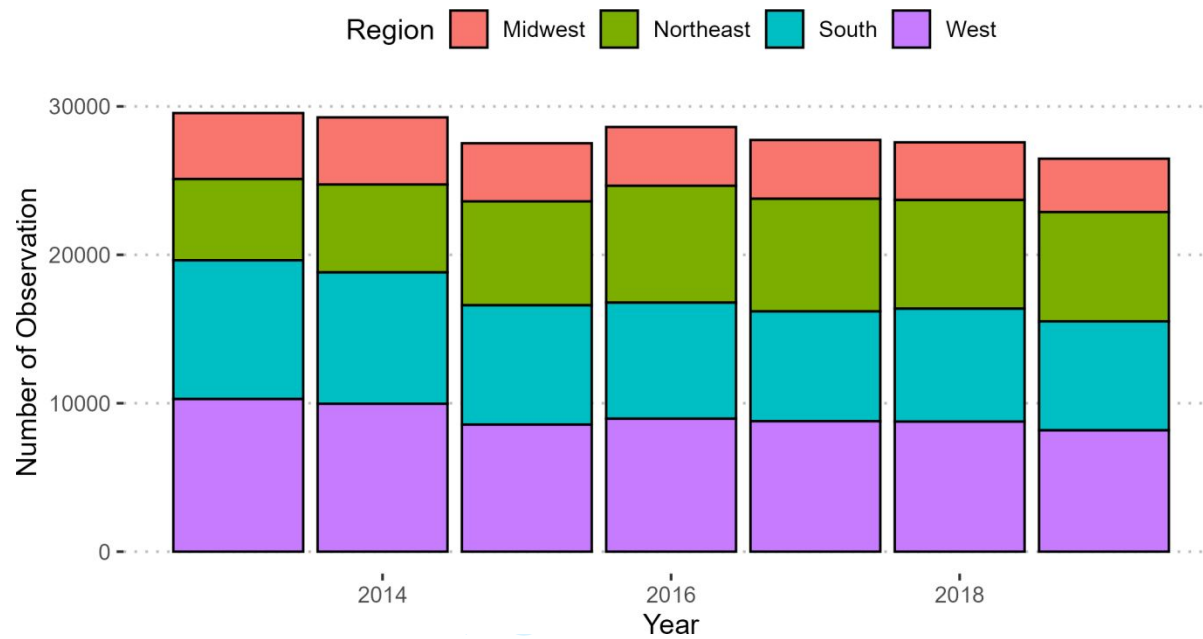


Figure A1: The number of observations by year in the analytical sample.

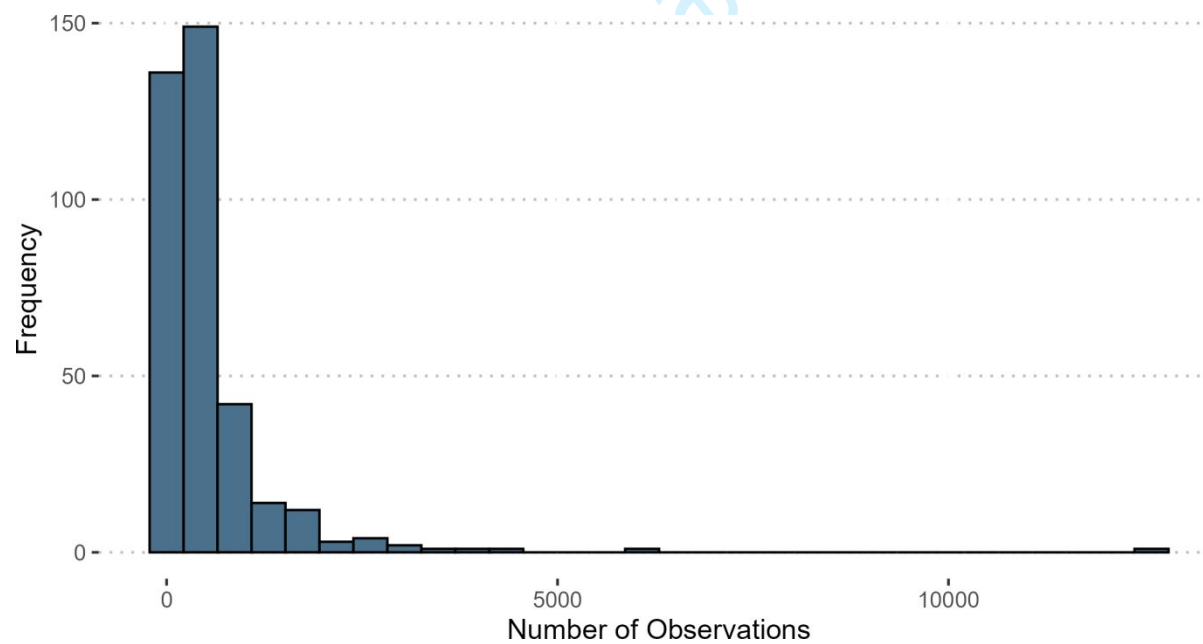


Figure A2: Distribution of counties' number of observations in the analytical sample.

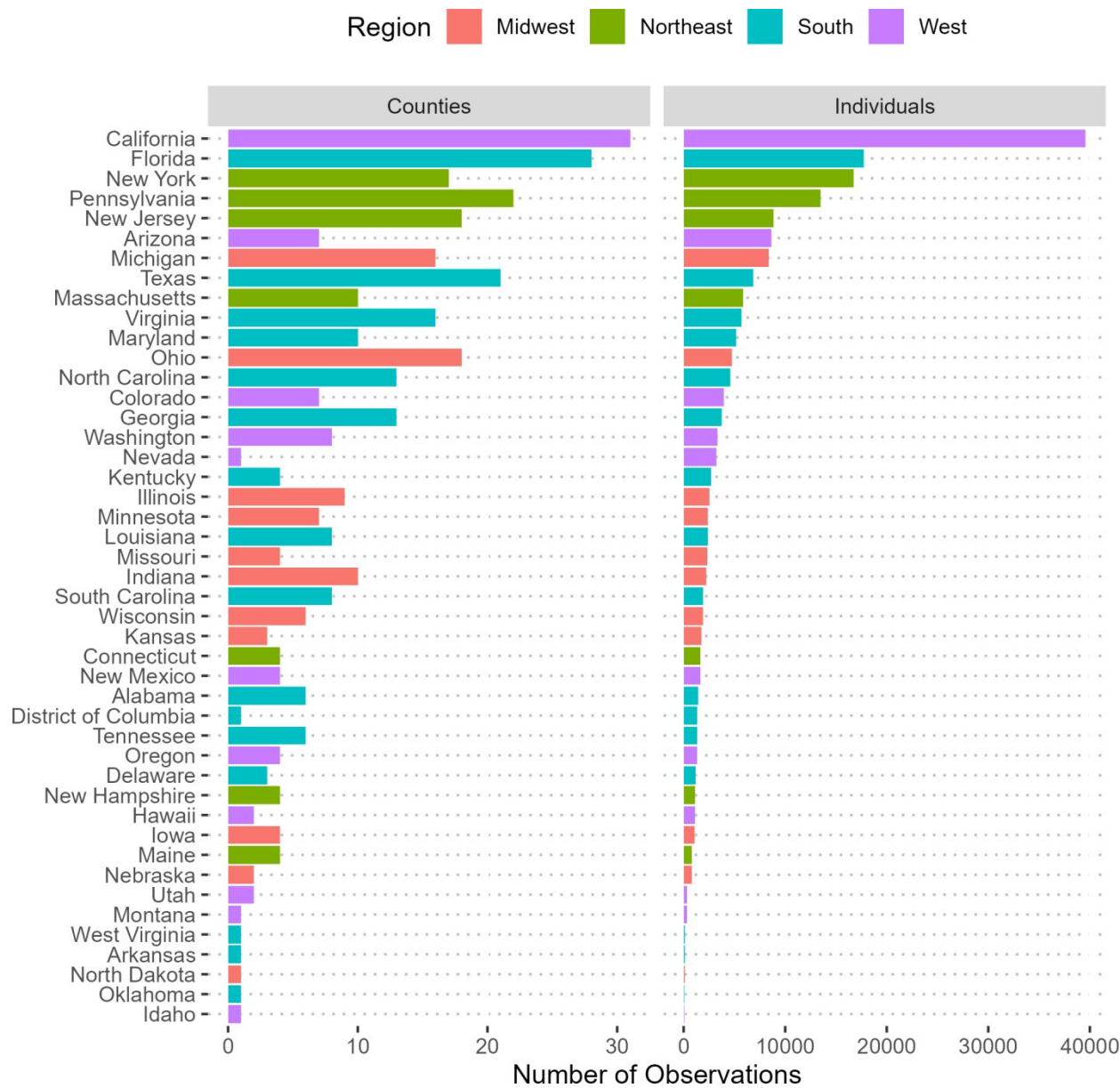


Figure A3: The number of individual and county observations by states in the analytical sample.

Appendix B. The Assumed Causal Structure

Figure B1 shows the assumed structure of causal pathways and the causal (green) and biasing (red) paths given the treatment (green), outcome (blue), and adjustments (white) specifications. With the hierarchical nature of the study, the top half of the DAG shows the group-level causal pathways and the bottom half shows the individual-level pathways. For both levels, the DAG assumes that there is a latent unit-specific feature - represented by nodes 'County-Year' and 'Individual' - that can be accounted for by observed factors and an unobserved component. These observed factors, furthermore, are assumed to depend on one another only through common causation from the latent feature without additional direct causal relationships among them - an unrealistic but inconsequential assumption so long as their relationships do not interfere with the focal causal relationship. The two levels are mutually causally connected to account for cross-level dependence.

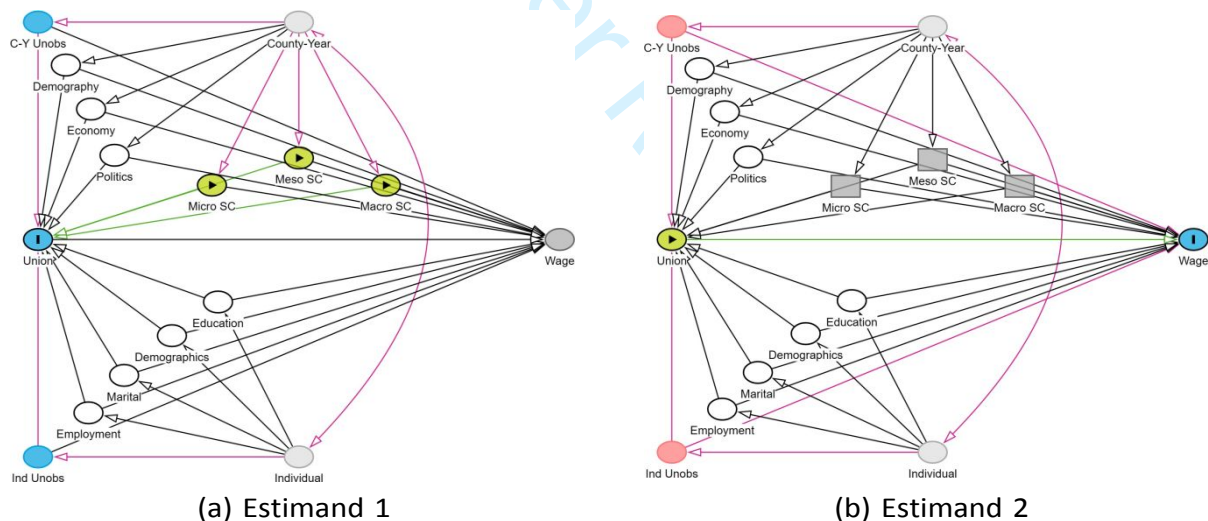


Figure B1: DAGs for Estimand 1 and 2

Basically, the DAG assumes that an individual's union membership has a causal effect on their wages, and this causal effect is confounded by individual- and group-level confounders. At the individual level, there are differential selections into union memberships *and* wage outcomes by characteristics such as demographics, education and marital status (Farber et al., 2021). Different industries, occupations and sectors also have different levels of union-

isation and wages. At the group level, the prevalence and potential for unionisation and the wage levels are affected by the local demographic-macroeconomic composition and political and labour market institutions. The three social capitals, as group-level variables, have the same causal paths as the group-level confounders.

Figure B1 shows that in the assumed causal structure, the adjustment specification blocks all biasing pathways except those going through unobserved confounders and does not induce collider or over-control bias.

The DAGs, however, could be incorrectly specified, thus leading to identification bias when using the adjustment specification described above. For example, the DAGs above assume that the political context in a county in a given year affects but is not affected by the levels of social capital and individuals' union membership and wage. However, it is also plausible for the political context to be affected by these variables or their unobserved ancestors, such as past outcomes. For example, trade unions may campaign for a certain political party or against legislation such as the RTW law and the levels of social capital may affect the effectiveness of such campaigns, thereby affecting the political context. Similarly, individuals' wages may also affect their voting behaviour, thus affecting the political context. In these cases, the political context becomes a collider and adjusting for it would introduce collider bias. To account for such situations, I estimate and report multiple models with stepwise addition of adjustment variables.

Appendix C. Variable Processing

Name	CPS Var. Name	Date Type	Note
Union Membership	unionmme	binary	
Hourly Earning	earnwke, uhouse	continuous	earnwke/uhouse
Age	age	continuous	
Sex	sex	categorical	
Hispanic	ethnic	binary	Assign 1 for all with non-zero response.
Race	race	categorical	Five categories: White (1), Black (2), American Indian (3), Asian (4), and Other (else).
White	-	binary	Assign 1 if race = 1 and Hispanic = 0
Education	edu	categorical	Five categories: Below High School (≤ 38), High School (39), Some College (40), Associate Degree (41&42), Undergraduate Degree (43), Postgraduate (≥ 44).
Public	class94	binary	Including employers at the Federal (1), state (2) and local (3) governments.
Occupation	docc00	categorical	
Industry	dind02	categorical	
Never married	marital	binary	Assign 1 if marital = 7

Table C1: Individual-level Variables from CPS